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9.1 Catalyst Monitoring

P0420/0430

9.1.1 Diagnostic Overview

The ECM tests the catalyst system during steady state driving by cycling the fueling LEAN and then RICH for a calibratable number of cycles while monitoring the oxygen storage capacity (OSC). Prior to the Catalyst test the canister purge valve is completely closed or completely opened with low canister purge value. This is to eliminate the influence of canister vapors on the downstream sensor during the test.

The first lean to rich cycle of the test is only used to establish an average voltage value of the downstream sensor voltage. During subsequent cycles the OSC is based on the integrated (accumulated) value of the difference between the average value of the previous lean to rich cycle and the measured instantaneous voltage during the current lean to rich cycle.

9.1.2 Monitoring Function

Catalyst monitoring is based on the monitoring of the oxygen storage capability by comparing the signals of the O2 sensor upstream and downstream the catalyst.

The engine control results in regular lambda oscillations of the exhaust gas. These oscillations are damped by the storage activity of the catalyst. The amplitude of the remaining lambda oscillations downstream the catalyst indicates the oxygen storage capability.

If all monitoring conditions are fulfilled, then a special defined A/F-modulation will be done. The relation of the deviations between the current downstream-sensor-signal to the average value of the downstream-sensor-signal is a sign for catalyst condition. The catalyst system is considered malfunctioning, if after a specified number of monitoring cycles the average of the ratios exceeds a threshold. The corresponding fault code is stored.



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9.1.3 Variables List

Siemens Parameter SAM/Specification	Description
CAT_DIAG	Result value of Cat diagnosis
SUM_CAT_DIAG	Counter Increment
CAT_DIAG_CLC	Current Cat Diagnosis
VLS_CAT_RATIO_SUM	Final Sum of values



9.2 Misfire Monitoring

P0300, 0301, 0302, 0303, 0304, 0305, 0306

9.2.1 Monitoring Function

The method of engine misfire detection is based on evaluating the engine speed fluctuations.

The engine torque is a function of engine speed, engine load and the moment of inertia.

In order to detect misfiring at any cylinder, the torque of each cylinder is evaluated by metering the time between two ignition events, which is a measure for the mean value of the speed of this angular segment. This means, a change of the engine torque results in a change of the engine speed.

It is also an influence of the load torque. This means, the influences of different road surface, e. g. pavement, pot holes etc. If the mean engine speed is measured, influences caused by road surfaces have to be eliminated.

This method consists of following main parts:

Data acquisition:

The duration of the crankshaft segments is measured continuously for every combustion cycle.

Sensor wheel adaptation:

Within a defined engine speed range and during fuel cut-off, the adaptation of the sensor wheel tolerances, instead of the misfire detection, is carried out.

With progressing adaptation the sensitivity of the misfire detection is increasing. The adaptation values are stored and taken into consideration for the calculation of the engine roughness.

Calculation of the engine roughness:

The engine roughness is derived from the differences of the segment durations. Different statistical methods are used to distinguish between normal changes of the segment duration and the changes due to misfiring.

Determination of misfiring:

Misfire detection is performed by comparing the engine roughness threshold value with the engine roughness value. If the threshold is exceeded, single misfire is detected.



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9.2.2 Statistics, Fault processing:

Emission Limit: (e) (3.2.2)

If the sum of cylinder(s) misfire counters within 1000 revolutions is 4 times exceeding a predetermined value during a driving cycle, or during the first 1000 revolutions, the fault code for emission relevant misfiring is temporary stored. If the following driving cycle is also above the emission limits, the MIL will be switched on and a cylinder selective or global fault will be stored.

Catalyst Damage: (e) (3.2.1)

If the weighted sum of cylinder(s) misfire counters within 200 revolutions is exceeding a predetermined value the fault code for catalyst damage relevant misfiring is stored and the cylinder with the highest rate will be switched off and the MIL will be switched on immediately. If two cylinders are switched off and the misfire rate is still above the damage limits, MIL is flashed immediately. If one of the cylinder selective counters is exceeding the predetermined threshold the following measures take place:

1. The Lambda closed loop system is switched to open-loop condition.
2. The cylinder individual fault code is stored or if multiple cylinders, then the Global fault code is set.
3. Fuel supply of the misfiring cylinder(s) is not cut-off (per customer request)
4. No downstream fuel trim.

All misfire counters are reset after each interval.



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9.2.3 Calibration Variables

Siemens VDO Parameter SAM/Specification	J1930 Description	Cal Value
C_MIS_B_MIN_NR	Threshold for CARB B detection	
C_MIS_A_IND_THD	Cylinder individual threshold used for cylinder shut-off during CARB A criterion window	
C_MIS_SUM_A_THD	Cylinder global threshold for misfire status CARB A detection	
THD_ER_AV	Engine speed fluctuation threshold for misfire detection	Table

9.2.4 Function Variables

Siemens Parameter SAM/Specification	VDO	J1930 Description
MIS_NR_TDC_ER		Number of misfire (for all cylinder)
MIS_SUM_ER(SEG)		Number of misfire (cylinder selective)
MIS_B_(SEG)		CARB B relevant misfire sums (cylinder individual)
MIS_CTR_A_(SEG)		CARB A relevant misfire sums (cylinder individual)
MIS_SUM_A		CARB A misfire weighted sum (global)

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9.3 EVAP-System Leak Diagnosis (Module DM-TL)

9.3.1 Evaporative system leak measurement

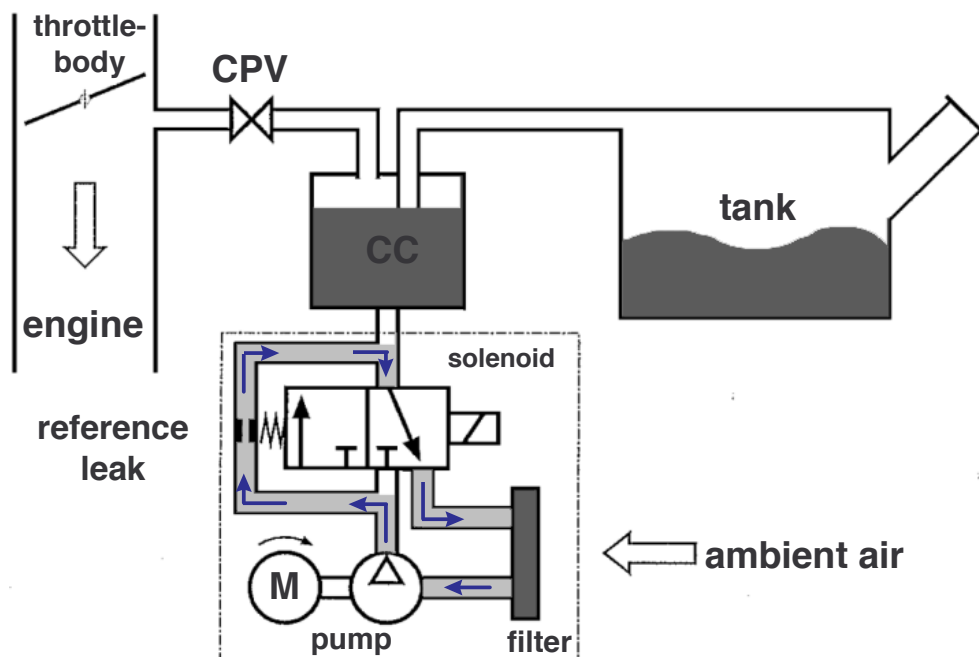
P0442/0456, 1434, 1447, 1448, 1449

9.3.1.1 General description of leak measurement

The evaporative system monitoring permits the detection of leaks in the evaporative system with a diameter of 0.02 inches and up.

By means of a Diagnostic Module Tank Leakage (DM-TL), an electrical actuated pump located at the atmospheric connection of the evaporative canister, a pressure test of the evaporative system is performed in the following order:

a) During the Reference Leak Measurement, the electrical actuated pump delivers through the reference restriction. The engine-management system measures the pump's electrical current consumption in this section.





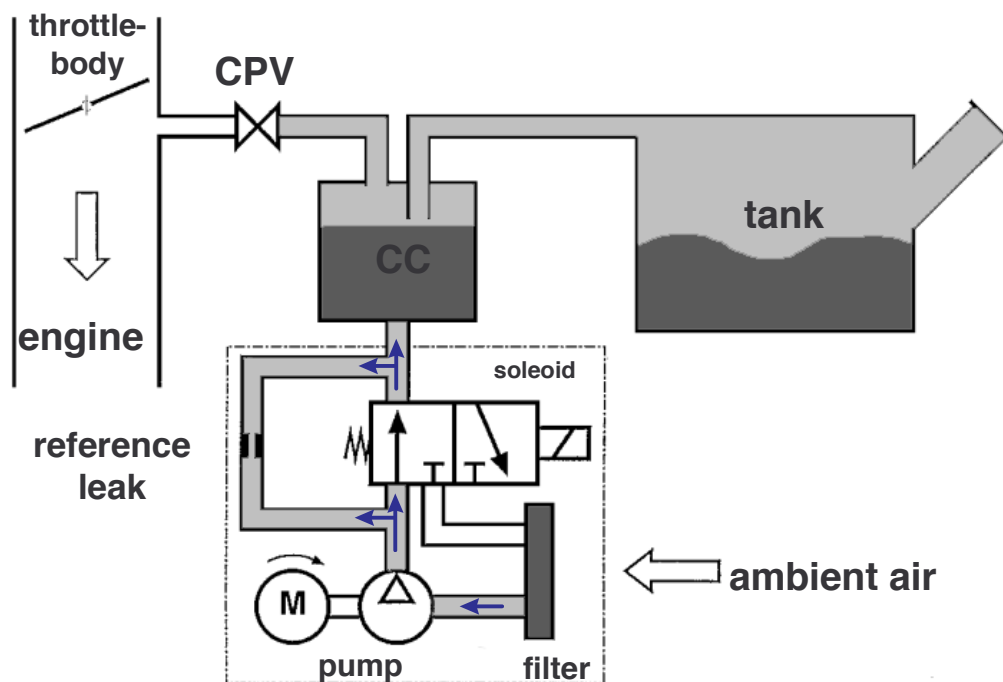
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b) During the Leak Measurement, the electrically actuated pump delivers through the charcoal canister into the fuel-tank system. The pressure in the evaporative system may be up to 2.5 kPa depending on the fuel level in the tank. The engine-management system measures the pump's electrical current consumption. A comparison of the currents of the reference leak measurement and the leak measurement is an indication of the leakage in the tank.



0.02 inch diagnosis diagnosis (P-Code 0456)::

The first step of the diagnosis is the reference measurement, the result of the pump reference current is stored (picture in chapter a). After the solenoid switches, the venting system is pressurized (picture in chapter b). In the rough leak measurement the rough leak threshold is reached, if the leak is smaller than 0,04 inch and then the small leak measurement phase follows. When the DMTL current reaches the reference current within the small leak time, the system is tight (leak smaller than 0,02 inch), otherwise a small leak between 0,02 – 0,04 inches is detected.



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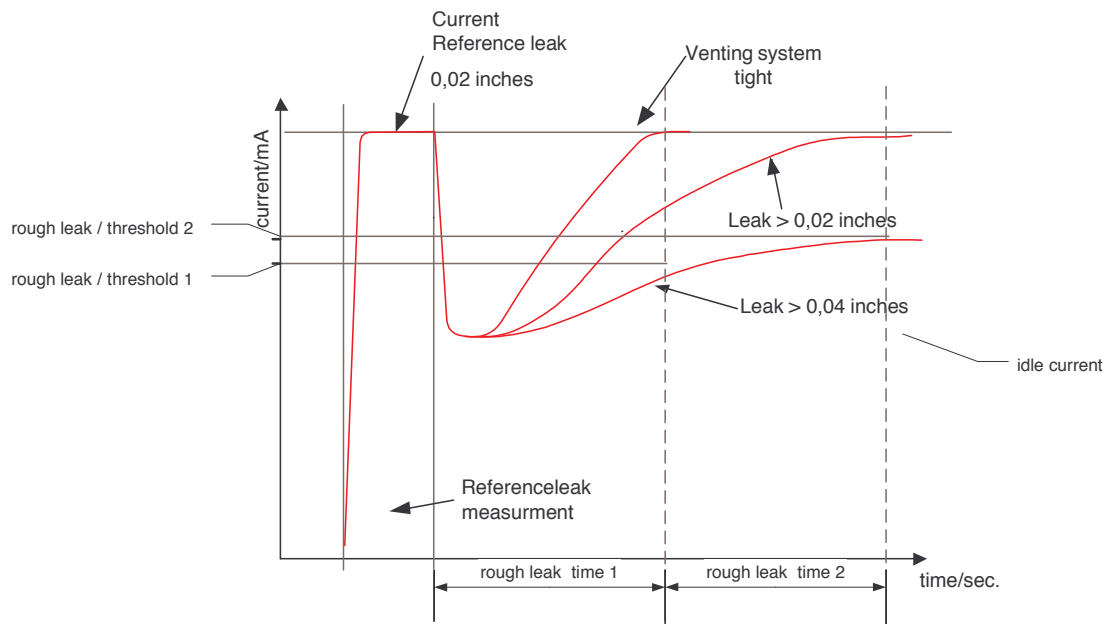
0.04 inch diagnosis (P-Code 0442):

The first step of the diagnosis is also the reference measurement, the result of the pump reference is stored (picture in chapter a). After the solenoid switches, the venting system is pressurized (picture in chapter b). In the rough leak phase (time) the pump current must reach the rough leak threshold 1 (rough leak threshold 1 = idle current pump + $K1 \times (\text{reference current} - \text{idle current})$). Factor $K1$ is between 0,16 and 0,28 depending on the characteristic current value of the pump (reference current – idle current), this value is various in every pump.

If the rough leak threshold 1 is not reached in the rough leak time, the rough leak threshold 2 must be reached in an additional time (rough leak threshold 2 = idle current pump + $K2 \times (\text{reference current} - \text{idle current})$). Factor $K2$ is between 0,60 and 0,80 depending on the characteristic current value of the pump (reference current – idle current).

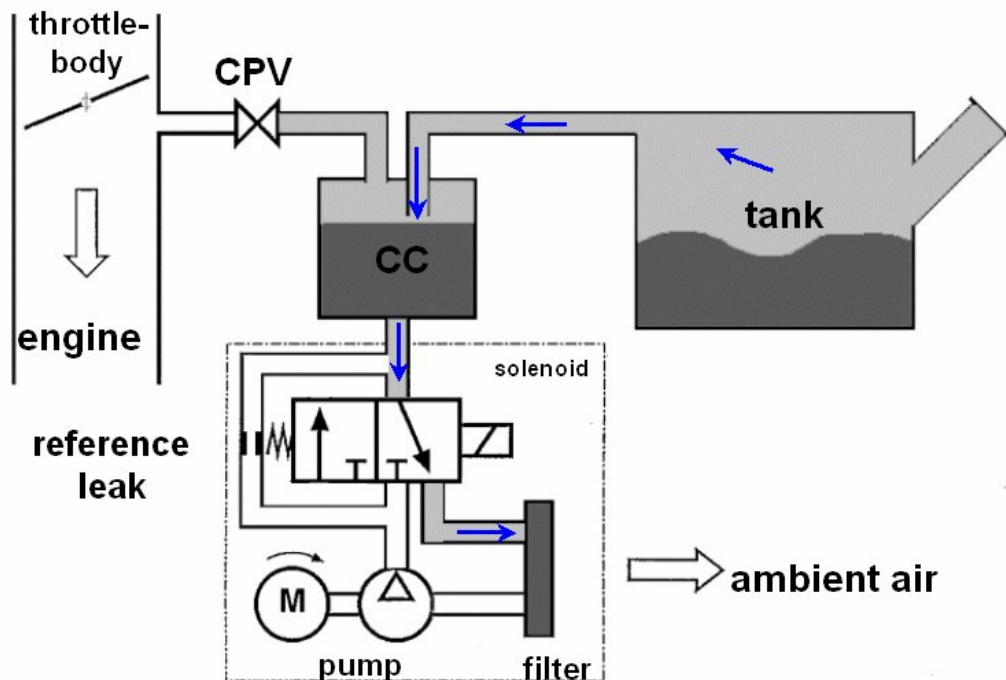
If the rough leak threshold 2 is also not reached, a leak > 0,04 inches is detected.

In the diagram below is the typical current of a tight system, a 0.02 inch leak, and a leak > 0.04 inches.





- c) After the test the remaining pressure in the evaporative system is bled off through the charcoal canister by switching off the pump and solenoid.



Description of DMTL fault codes (P 1434, 1447, 1448, 1449):

In the reference measurement phase the current of the DMTL is checked, if the current consumption is smaller than 15 mA a defect pump/motor is detected, and P1448 is stored . If a current consumption of bigger than 40 mA is measured also a defect pump/motor is detected, and the P1449 is stored.

After the reference measurement the electrical solenoid is switched and the venting system is tight. The solenoid is checked by comparison of the reference currence and the idle current immedatetely after the solenoid has switched.

If the difference between the two currences (reference currence – idle currence) > 2 mA is not reached, the fault code 1447 is stored.

The P 1434 fault code is set, if current fluctuations (caused by humidity in the DMTL) > 1mA are detected.



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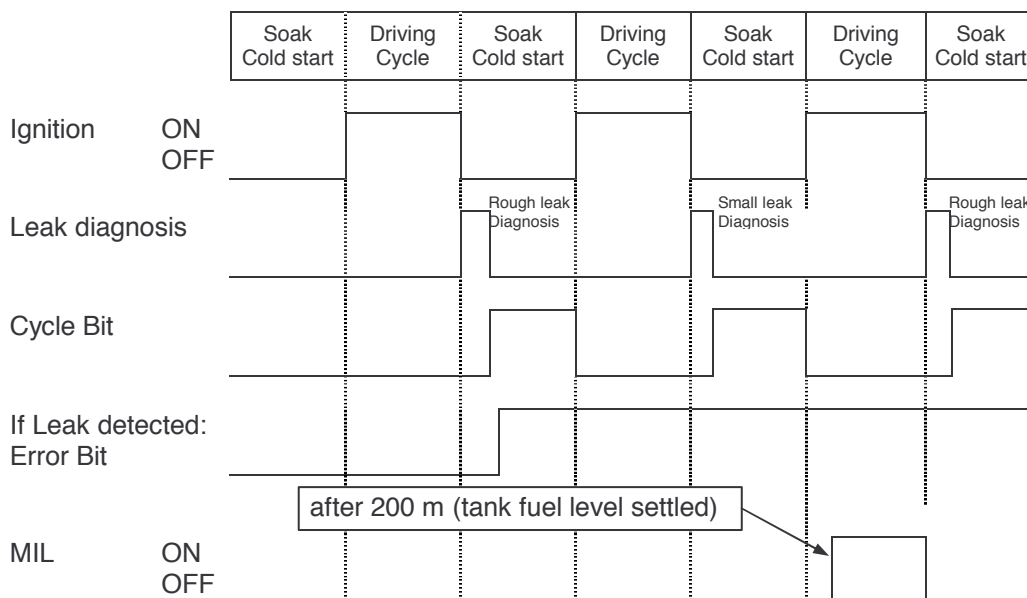
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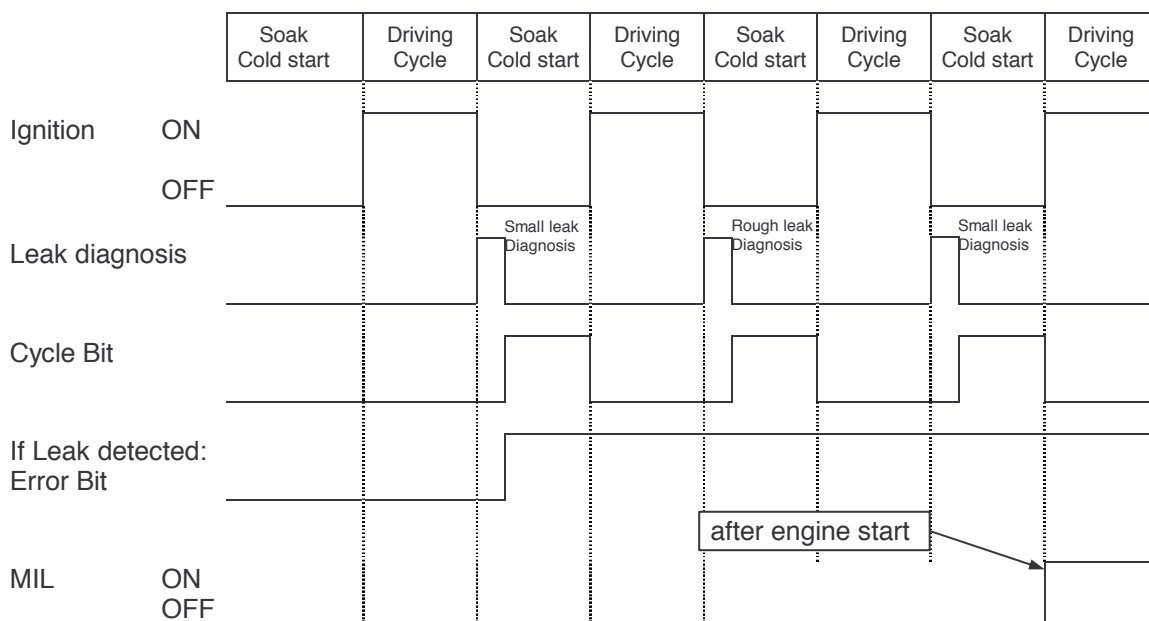
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9.3.1.2 Diagnosis Frequency and MIL illumination

**9.3.1.2.1 Diagnosis Frequency and MIL illumination -
no refueling detected, leak > 0.04 inches**



**9.3.1.2.2 Diagnosis Frequency and MIL illumination -
after refueling detected, leak > 0.02 inches**





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9.3.2 EVAP (Functional check canister purge solenoid)

P0440

9.3.2.1 Monitoring the canister purge solenoid:

The diagnosis is used for the functional test of the CP solenoid (CPS).
The test consists of three steps. Step 2 can be triggered separately by an EOL (End Of Line) request by a serial communication tool.

Step 1: based on the activated charcoal filter (ACF) load degree

Step 2: based on . . .

- the engine speed change at idle speed
- the deviation of lambda-controller
- manifold air pressure (MAP) change at idle speed

Step 3: based on the difference between the measured air mass flow before
and during a CPS opening

9.3.2.2 Monitoring function

The first check of the CPS is based on the ACF- load degree. The "Canister Load diagnosis" is calculated permanently until the complete check CPS is finished.

During the next check, the CPS is evaluated based on the engine speed change at idle speed. To this effect, the CPS is opened for a short time and the engine speed monitored for a certain period.

After this check has been enabled for the first time, it is requested during each idle speed phase. This is repeated as long as a result has been reached. This check is not bound to one IS phase, but can be distributed to several IS phases.

During this check the lambda deviation will also be checked.

In step 3 the CPS is considered on the basis of the measured mass air flow before and during a CPS opening phase.

If the CPS is detected to be not OK after all three checks have been passed (end of step 3), then the error is set.



9.4 Fuel Level Sensor

P0461: Plausibility error

P0462: electrical short circuit ground right

P0463: electrical open circuit right

P2067: electrical short circuit ground left

P2068: electrical open circuit left

The diagnosis of the fuel level sensor signal consists of a circuit continuity check and a rationality check.

9.4.1 Fuel level sensor circuit continuity check

The signal of the fuel level sensor is monitored concerning the valid range. This range depends on the used fuel level sensor.

If the fuel level sensor signal is out of the valid range, an electrical malfunction (short circuit to ground or interruption) is detected and an appropriate fault code is set.

If the left or right fuel level sensor signal is above the upper threshold, a short circuit plus is detected and an appropriate fault code for the left or right sensor is set. If the left or right fuel level sensor signal is below the lower threshold, a short circuit ground is detected and an appropriate fault code for the left or right sensor is set.

9.4.2 Fuel level sensor signal rationality check (plausibility error)

The engine management system of every BMW has the capability to calculate fuel consumption. For the fuel level sensor plausibility check, this calculated consumption is compared with the difference of the fuel level signal. When the calculated fuel consumption reaches an appropriate and predetermined amount (for example five gallons), the calculated fuel consumption is compared to the change in fuel level as indicated by the fuel level sensors. If the difference is greater than the applicable threshold value, a stuck fuel level sensor fault is detected and an appropriate fault code is set.

If a fault is present, the OBD II EVAP leak monitor will run using a substitute value of 85% total fuel tank volume.

The 85% substitute value will assure that in every case the required 0.020 inch leak is detected by the OBD II system.



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9.4.3 FLS diagnosis frequency of FLS circuit continuity check
9.4.3.1 short circuit battery

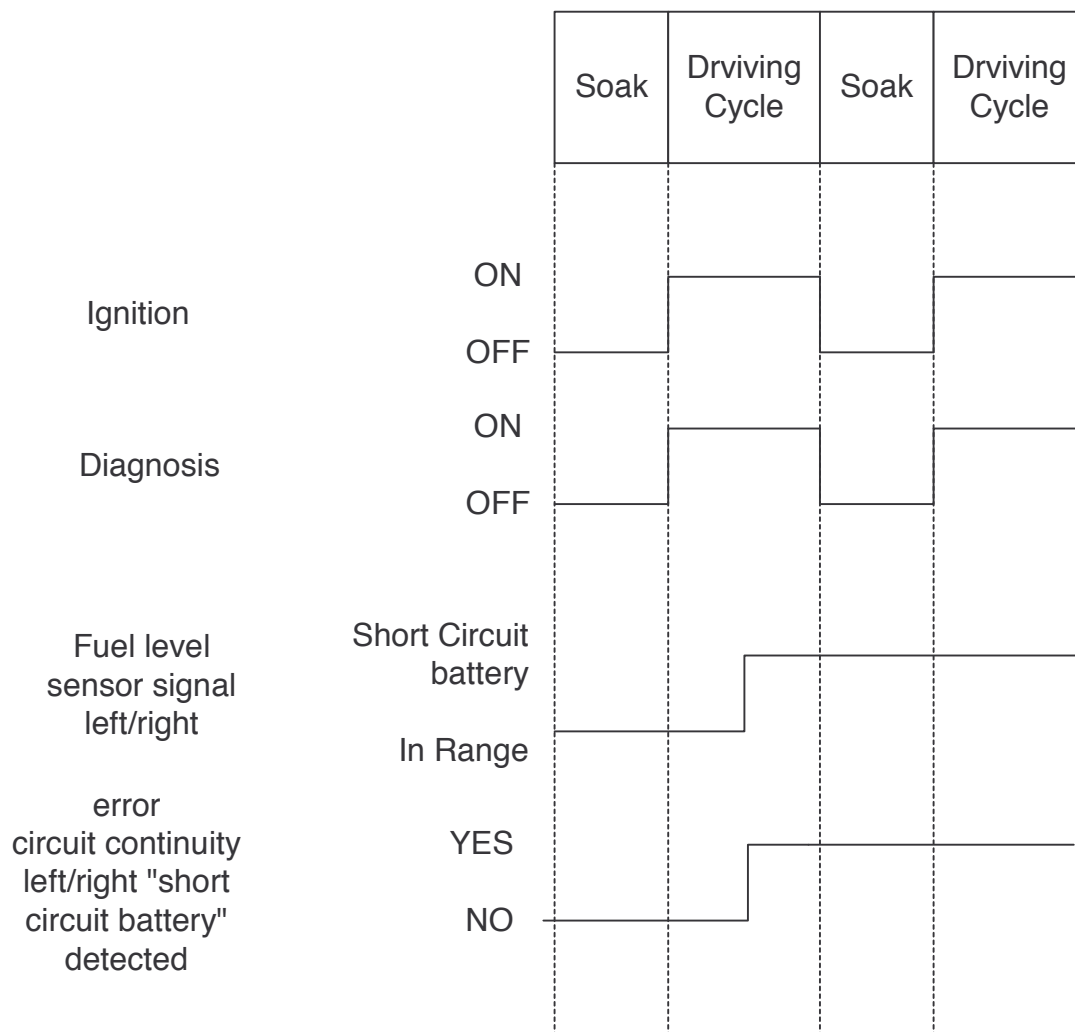


Figure 1



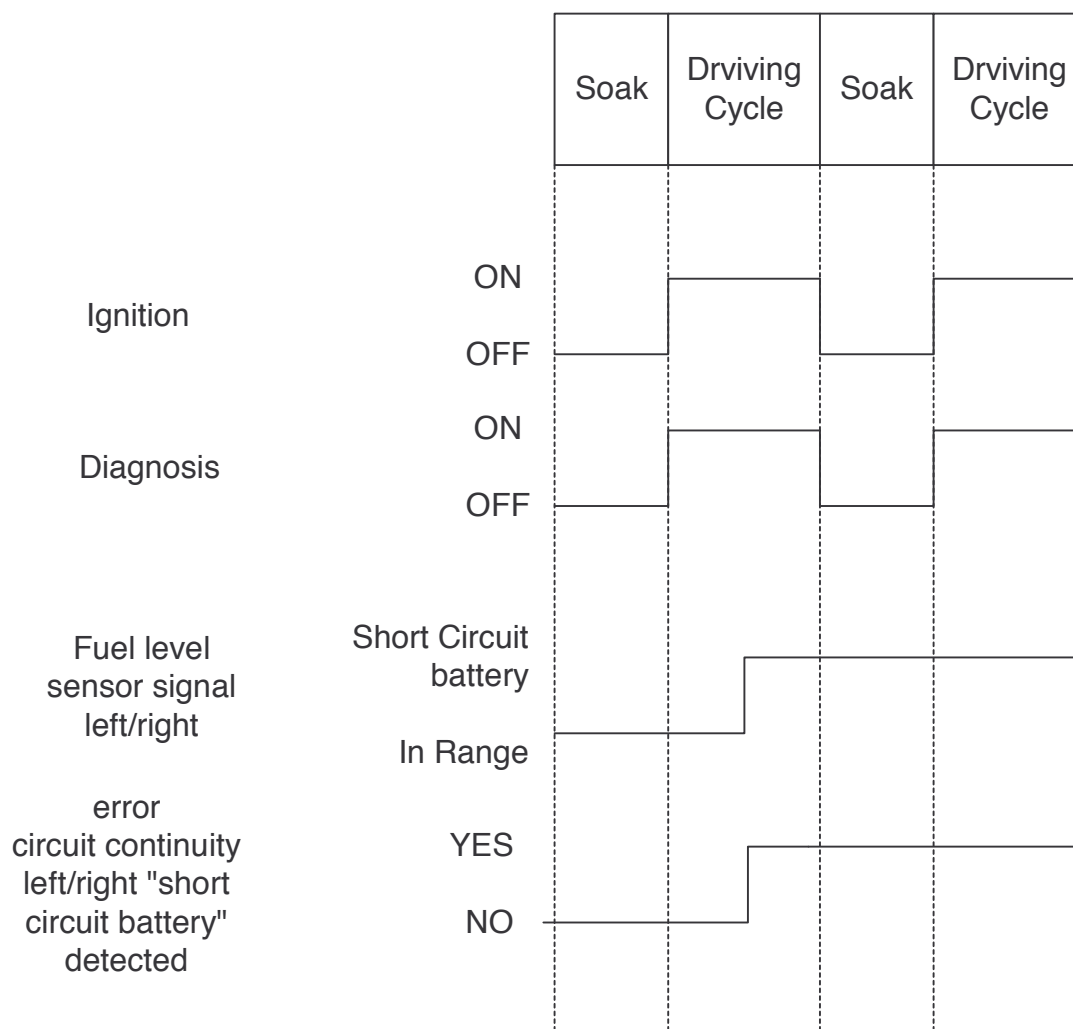
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9.4.3.2 short circuit ground





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9.4.4 FLS diagnosis frequency of FLS rationality check (plausibility error)

		Soak	Driving Cycle	Soak	Driving Cycle
Ignition	ON				
	OFF				
Diagnosis	ON				
	OFF				
Engine consumption counter >5 gal.	YES				
	NO				
difference between engine consumption counter and fuel level change < -3.9 gal. or > 3.9 gal.	YES				
	NO				
error fuel level plausibility detected	YES				
	NO				



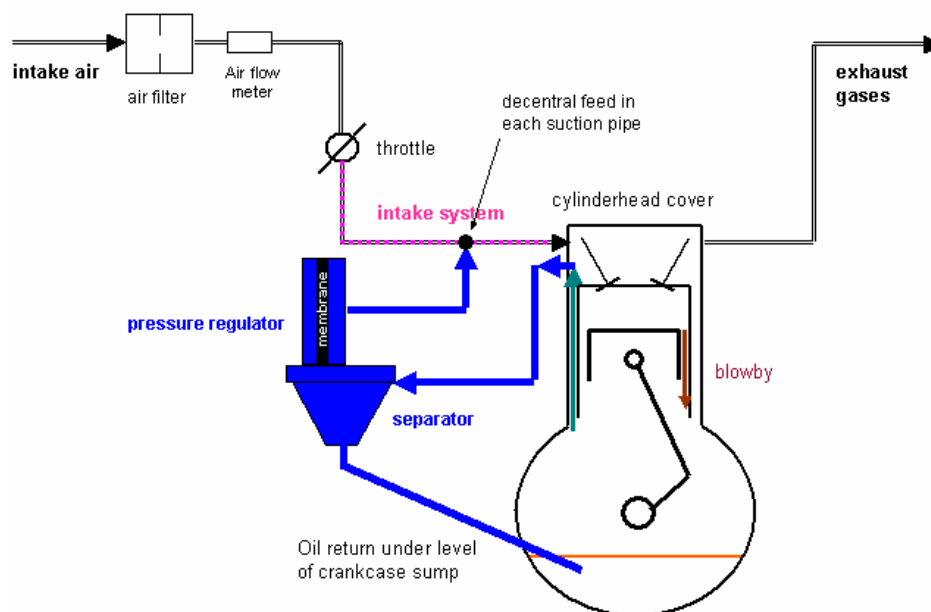
9.5 Positive Crankcase Ventilation (PCV) System

9.5.1 General description of the PCV-System:

There are 3 tubes connected to the engine: The first of them conducts the blow by gases from the cylinder head cover to the separator, where the oil is separated from the air and lead back by a second tube to the crankcase sump. A third tube directs the cleaned blow by gases via the intake system to the combustion. The pressure regulator makes sure that the high vacuum level between crankcase and ambient air will be reduced if needed.

9.5.2 Diagnosis of a leakage in the PCV-System:

A disconnection or leakage in the PCV-System is indicated by a rough or stalling engine and results in a reaction within the fuel system (fuel trim deviation). In this case a fault code will be stored by the fuel system monitoring.



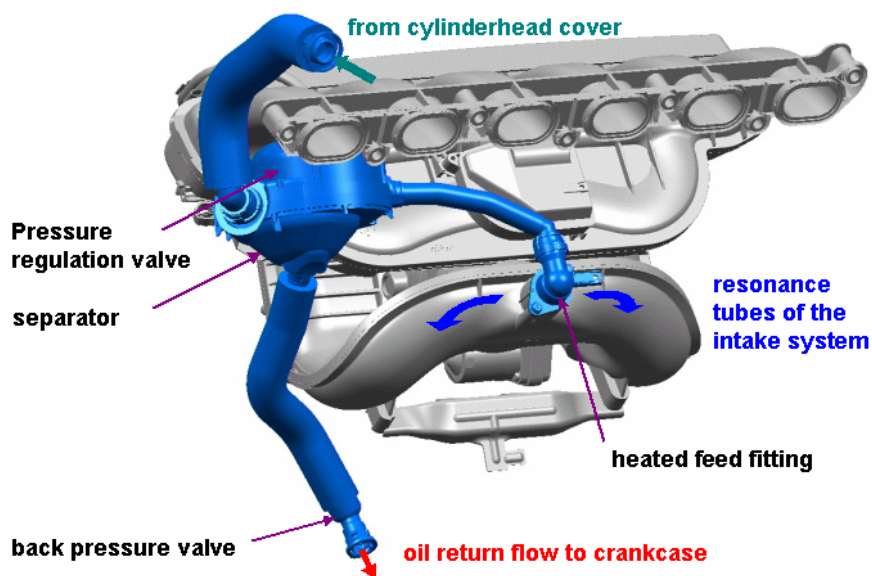


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9.6 Ambient Temperature Monitoring

9.6.1 CAN based Ambient Temperature – signal diagnosis *P0072, P0073*

9.6.1.1 General Description

The purpose of this diagnosis is to detect electrical faults as defined in OBDI requirements. The input signal is a CAN message of instrumentation cluster. If an error is present on CAN signal, an error symptom is set and an error counter is de-bounced.

9.6.2 Ambient Temperature Signal Plausibility Check *P0071*

9.6.2.1 General Description

This diagnosis is performed in order to detect a stuck or not plausible TAM signal which cannot be detected by electrical range diagnosis.

The first part, just after start looks on the change of ambient temperature and compares the start and stop temperature. If the check is positive the diagnosis is finished. In negative case diagnosis runs to next step during warm up phase.

The error detection is only performed if the monitoring conditions for time after start, engine state idle speed, time of engine stop, ECT and ambient temperature are fulfilled. The plausibility error is detected if the absolute value of the temperature-difference between the arithmetic mean of ECT and TIA and the ambient temperature TAM (in formula: $ABS[(ECT+TIA) \times 0,5 - TAM]$) exceeds the threshold for an anti-bounce time.

The error validation is only performed if ECT signal is valid and the vehicle was driven with a certain vehicle speed. If both conditions are true and an error was detected, then the error is set for this driving cycle and the diagnosis is switched off.

For RBM handling the Cold Start Denominator will be considered.



9.7 Fuel System Monitoring

9.7.1 Lambda Adaptation

9.7.1.1 Diagnostic Overview / Monitoring Function:

Sec (e) (6 – 6.4.6) (1968.2)
(P0171, 172, 174, 175)

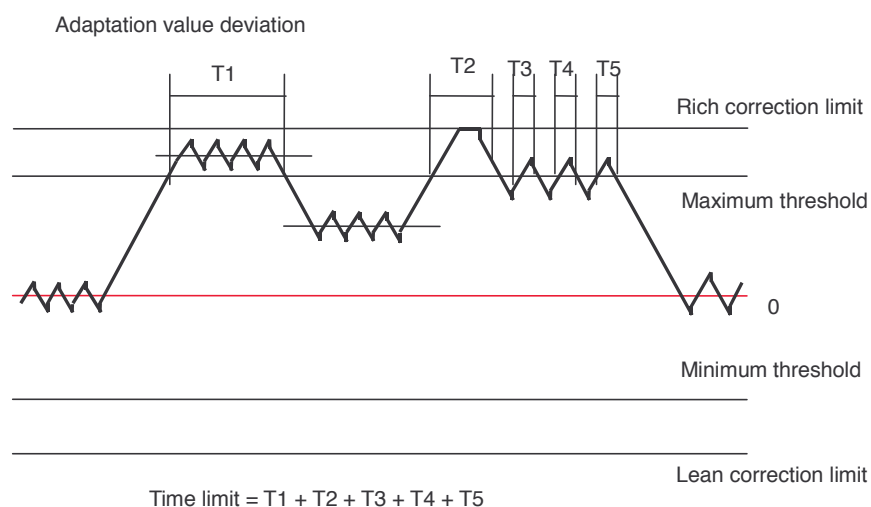
The ECM monitors the fuel system control continuously during all engine states except PUC (decel fuel cut-off). After the enable conditions are met a counter is started. At this point the ECM evaluates the total percentage of short and long term fuel control. If no condition is present the end diagnostic counter will decrement from a calibratable value to zero and a passing decision is made.

If a lean condition is present and total fuel control is above the calibratable threshold two timers are started. If the lean threshold counter exceeds the calibratable threshold before the reset timer has decremented from calibratable threshold to zero a lean error is set.

If a rich condition is present and total fuel control is below the calibratable threshold two timers are started. If the rich threshold counter exceeds the calibratable threshold before the reset timer has decremented from a calibratable threshold to zero a rich error is set.

The time counter is increased while “lambda controller + lambda adaptation” exceed minimum or maximum threshold.

The error is detected as soon as the time counter reaches its maximum value.





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9.7.1.2 Similar Conditions Function

Sec (e) (6.4.2)

When the engine management system recognizes a failure in the misfire or fuel systems, the engine management system is required to record the conditions present when the fault occurred. These conditions recorded include engine speed, engine load (MAF), and warm up status of the first event that resulted in the storage of a code. These conditions stored are referred to as **similar conditions**

Once the similar conditions are meant without a failure in the misfire or fuel system, the flag is set to 1. Once this flag is set the driving cycle counter for that failure can be decrement.

The code and stored freeze frame conditions may be erased if similar conditions are not encountered during the next 80 driving cycles immediately following the initial detection of the malfunction.

The MIL may be extinguished after three sequential driving cycles in which similar conditions have been encountered without an exceeding the thresholds of the fuel system diagnostic.



9.7.2 Trim Control Plausibility Monitoring *P2096/2098, P2097/2099*

9.7.2.1 Monitoring function:

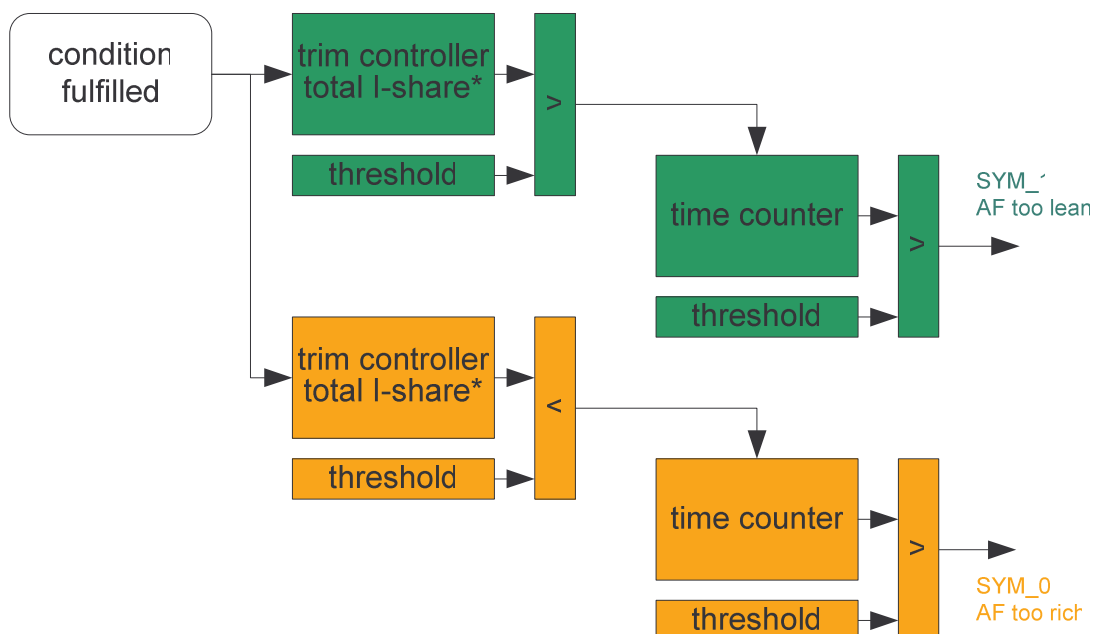
The trim control plausibility monitoring detects a high deviation of the I-share of lambda trim control. If it exceeds given thresholds the following malfunction is detected:

Fuel trim above limit

If the above mentioned malfunction is detected, the corresponding fault code is stored.

	B1S1	B2S1
Air fuel mixture too rich	P2097	P2099
Air fuel mixture too lean	P2096	P2098

9.7.2.2 Block diagram:





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9.8 Oxygen Sensor Monitoring linear upstream / binary downstream

9.8.1 Upstream Oxygen Sensor Monitoring (linear)

9.8.1.1 Upstream Oxygen Sensor - Short Circuit Monitoring

P0131/0151, P0132/0152

9.8.1.1.1 Monitoring function:

The oxygen sensor circuit monitoring detects the following malfunctions by evaluating the error information received from oxygen sensor microcontroller:

- Short circuit of sensor signal to battery voltage
- Short circuit of sensor signal to ECM ground

If one of the above mentioned malfunctions is detected, the corresponding fault code is stored.

	B1S1	B2S1
SC Ground	P0131	P0151
SC Vbatt	P0132	P0152



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9.8.1.2 Upstream Oxygen Sensor - Open Circuit Monitoring

P112C/112D, P2243/2247, P2626/2629

9.8.1.2.1 Monitoring function:

The oxygen sensor circuit monitoring detects the following malfunctions by evaluating the error information received from oxygen sensor monitoring functions:

	B1S1	B2S1
- Reference voltage Failure – (UN)	P2243	P2247
- Virtual Ground Failure – (VM)	P112C	P112D
- Pumping Current Failure – (IP)	P112C	P112D
- Trim Current Failure – (IA)	P2626	P2629

If one of the above mentioned malfunctions is detected, the corresponding fault code is stored.

9.8.1.2.2 Monitoring Description:

This function determines, if an open circuit in any of the four electric lines (*Reference Voltage, Virtual Ground, Pumping Current and Trim Current*) is present in the WRAF Sensor.

This function shall be triggered only if one of the following diagnosis is active (to set the readiness bit), which are 'Upstream Oxygen Sensor - Signal Monitoring During Fuel Cut-off' and 'Upstream Oxygen Sensor - Heater Monitoring'. The function shall go to the state = "active" only if one of the above diagnosis detected a fault.

(Reference Voltage)

If a heater error exist and sensor voltage ist too low, while the internal resistance measurement is turned off, an open circuit in the line 'Reference voltage' occurred.

(Virtual Ground) or (Pumping Current)

An open circuit in line Virtual Ground or in the line Pumping current can be detected if the sensor signal stocks near lambda 1. The sensor non-activity can be detected by the Oxygen Sensor Signal Monitoring During Fuel Cut-off (signal voltage below e.g. 2,1 V *) in fuel cut off).

(Trim Current) If the sensor shows an augmented gain, i.e. the sensor signal is higher than the nominal characteristic line, the plausibility test during the fuel cutoff phase shall detect this symptom (signal voltage above e.g. 5,6 V *) during fuel cut off) and an Open Circuit is assigned to the line Trim Current.

)* For exact values please have a look at the summary table!



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9.8.1.3 Upstream Oxygen Sensor - Signal Controller Monitoring

P3022/3023, P3024/3025

9.8.1.3.1 Diagnostic Overview:

Sec (e) (7.2)

This function will detect an error during the initialization and/or operation of a WRAF sensor controller through SPI communication. Information communicated from the Basic Software (BSW) is used for initialization and communication between application software (ASW) and the controller. This is used to determine if the function is working properly.

9.8.1.3.2 Monitoring Function:

Sec (e) (7.2.1)

After an ECU reset, the WRAF sensor controller is started and the diagnosis determines the time until the initialization, has been performed in the allowed time. If not successful, then a DTC will be stored. If this is successful, then the difference is checked between the present error counter and the stored value of this error counter at ECU reset, (switching from Key "OFF" to Key "ON") or at clearing error memory and after each function call, in case a difference between both counters was found. If there is a difference, another counter is incremented. If this counter is higher than a threshold, a SPI communication error is stored.

	B1S1	B2S1
Communication error	P3022	P3023
Initialization error	P3024	P3025

All of the above checks are performed internal to the ECU. (See flowchart)



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9.8.1.4 Upstream Oxygen Sensor - Signal Activity Check
P2414/2415

9.8.1.4.1 Monitoring Description:
Sec (e) (7.1.1)

The oxygen sensor signal activity check monitors if the sensor is attached to the exhaust pipe and whether the exhaust is sampled correctly (no leakage). A malfunction is detected if the oxygen sensor voltage is above a threshold (shows too lean mixture in part load or full load)

If the above mentioned malfunction is detected, the corresponding fault code is stored.

B1S1 P2414
B2S1 P2415

9.8.1.5 Upstream Oxygen Sensor - Swapped Sensors Check
P0040

9.8.1.5.1 Monitoring Description:

This function will detect if the Oxygen Sensor wire harness has been cross connected, i.e., Bank 1 with Bank 2. This is performed by the use of the output of the fuel correction (lambda controller) of each bank. If this control is on opposite limit at bank 1 and bank 2, the sensors are swapped and the corresponding fault code is stored.

Corresponding fault code:

P0040



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9.8.1.6 Upstream Oxygen Sensor - Active Signal Check (Shift to lean / rich)

P2195, 2196, 2197, 2198

9.8.1.6.1 Monitoring Function

Sec (e) (7.1.1) (e) (6.2)

This monitor is an enhancement of the Downstream Oxygen Sensor - Active Signal Check (Stuck lean / rich) and the Trim Control Plausibility Monitoring. Its purpose is to help determine the root cause of a stocking downstream sensor signal or a implausible high i- share of the fuel trim controller .

The monitor will only be enabled if a fuel correction fault has been detected and a malfunction code has been stored (P2096 – P2097 – P2098 – P2099)

OR

if the rear sensor signal check has detected, that the rear sensor signal is very rich or very lean and the corresponding malfunction fault code has stored (P114A – P114B – PP114C – P114C)

If one of the listed fault codes exists, this diagnosis will be enabled to determine if the root cause of the malfunction is due to a stuck signal or characteristic line shift of the upstream O2 sensor or due to a stuck signal of the downstream O2 sensor or a system malfunction, i.e. vacuum leak, injector, etc...

If it has been determined that the upstream O2 signal was the root cause of the fuel correction fault, the appropriate DTC will be stored along with the fuel correction or with the downstream sensor stuck DTC.



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9.8.1.7 Upstream Oxygen Sensor - Signal Dynamic Monitoring (Slow Response)

P0133/0153

9.8.1.7.1 Monitoring function:

Sec (e) (7.3.1)

The oxygen sensor signal dynamic monitoring detects greater deviations of the dynamic behavior of the sensor signal compared to the nominal behavior, controlled by the lambda controller

The change of the dynamic behavior is caused by problems of the electrical connection (e.g. open circuit), extreme aging of the sensor or a low sensor temperature which slows down the sensor compared to the nominal behavior.

The monitoring is based on an amplitude criterion, i.e. the relation between the amplification of the oxygen sensor and the model is monitored and detects the following malfunction:

Sensor signal too slow

If the above mentioned malfunction is detected, the corresponding fault code is stored.

B1S1	P0133
B2S1	P0153



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9.8.1.8 Upstream Oxygen Sensor - Signal Monitoring During Fuel Cut-off

P2297/2298

9.8.1.8.1 Monitoring function

The oxygen sensor signal monitoring during fuel cut-off detects if the oxygen sensor signal is not plausible during fuel cut-off. A malfunction is detected if the oxygen sensor voltage is outside the normal operating voltage range (3,1 V ... 5,68 V).)*

If the oxygen sensor signal voltage is between 2,1 ... 3,1 V during fuel cut-off, then the signal is not plausible.

If the above mentioned malfunction is detected, the corresponding fault code is stored.

B1S1 P2297

B2S1 P2298

If the oxygen sensor signal voltage is above a threshold during fuel cut-off (e.g. 5,6 V)*) or below a threshold (e.g. 2,1 V)*) then the open circuit diagnostic function is triggered (see chapter 'Oxygen Sensor Monitoring - Open Circuit'). The fault processing continues in this function.

)* For exact values please have a look at the summary table!



9.8.1.9 Upstream Oxygen Sensor - Heater Monitoring

9.8.1.9.1 Diagnostic Overview:

Sec (e) (7.2.3)

The purpose of this function is to detect oxygen sensor heater failures that would lead to an increase in emissions beyond the thresholds stated in the appropriate regulations.

The diagnosis shall be carried out by determining whether the measured oxygen sensor ceramic temperature falls below set limits over a number of measurement cycles. The evaluations of the diagnosis cycle are determined after the completion of a limited number of monitoring cycles.

Deviations in the oxygen sensor ceramic temperature or the oxygen sensor not being operatively ready in a timely manner (because of a too low temperature) can lead to an increase in emissions above the applicable standards or prevent the sensor signal from being used as a diagnostic system monitoring device. Deviations may occur due to, for example, ageing of the heater element, defective wiring, increased heater circuit connector contact resistance, defective heater driver etc.

9.8.1.9.2 Monitoring Function:

The diagnosis strategy is based on a statistical evaluation of the oxygen sensor ceramic temperature over a pre-defined number of monitoring cycles. The oxygen sensor ceramic temperature shall be obtained indirectly via the measured internal resistance of the sensor.

If the sensor is not ready after a defined time (e.g. 30s after start))* the sensor is set to forced readiness mode and the Upstream Oxygen Sensor Heater Monitoring is started.

Two cases can appear:

- sensor temperature is invalid (no measurement of sensor temperature possible because of an ECU internal (electrical) failure)
=> P165F/P166F is stored
- sensor temperature is below a threshold
=> normal failure detection

A low sensor temperature can be caused by a weak heater or a open circuit in the temperature measurement line (line UN). After a low sensor temperature has been detected, the open circuit diagnosis is triggered to check, if an open circuit in line UN is present. If there is an open circuit, then open circuit fault code (P2243/P2247) is stored (see chapter 'Oxygen Sensor Monitoring - Open Circuit' and picture below).



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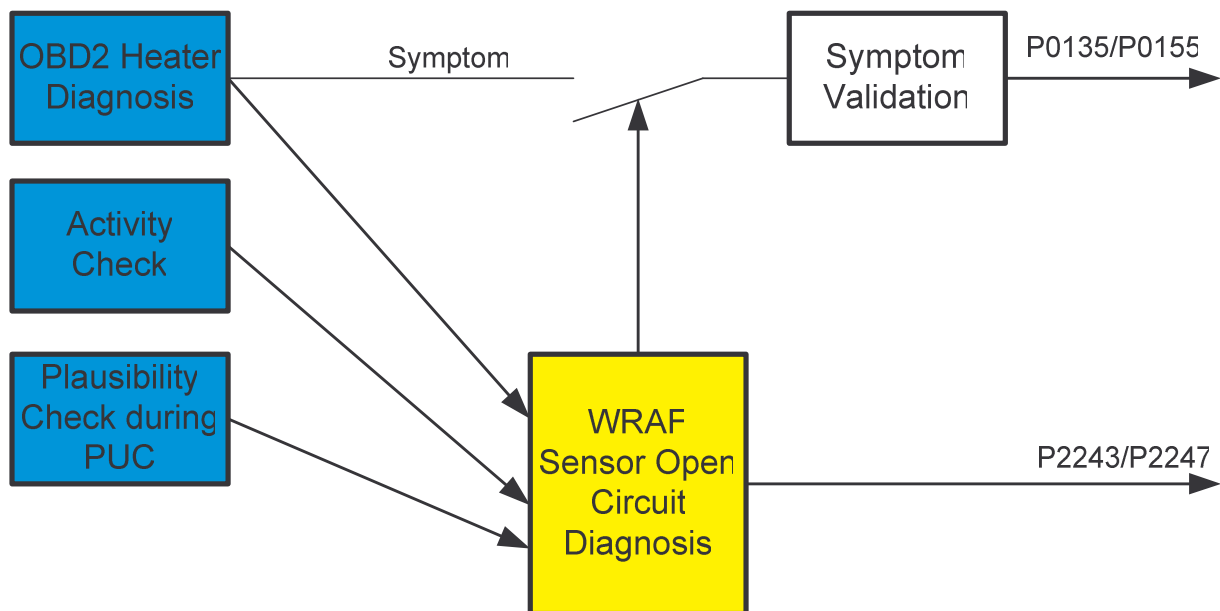
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If there is no open circuit present, then the heater fault code is stored (P0135/P0155).

)* For exact values please have a look at the summary table!

	heater power too low	sensor temperature invalid
B1S1	P0135	P165F
B2S1	P0155	P166F





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9.8.1.10 Upstream Oxygen Sensor - Heater Circuit Monitoring
P0030,0031,0032,0050,0051,0052

9.8.1.10.1 Monitoring function:

Sec (e) (7.3.3)

The oxygen sensor heater circuit monitoring detects the following malfunctions by evaluating the error information received from the power stage:

- HO2S up SCVB
- HO2S up SCG
- HO2S up Open circuit

If one of the above mentioned malfunctions is detected, the corresponding fault code is stored.

	B1S1	B2S1
SC Ground	P0031	P0051
SC Vbatt	P0032	P0052
Open circuit	P0030	P0050



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9.8.2 Downstream Oxygen Sensor Monitoring (binary)

9.8.2.1 Downstream Oxygen Sensor - Circuit Monitoring

P0137,0138,0140,0157,0158,0160

9.8.2.1.1 Monitoring function:

The oxygen sensor electrical monitor detects the following malfunctions:

- HO2S Down signal SCVB
- HO2S Down signal SCG
- HO2S Down Signal Open Line

If one of the above mentioned malfunctions is detected, the corresponding fault code is stored.

	B1S2	B2S2
SC Ground	P0137	P0157
SC Vbatt	P0138	P0158
Open circuit	P0140	P0160



9.8.2.2 Downstream Oxygen Sensor - Signal Dynamic / Plausibility Check During Fuel Cut-off
P0139/0159

9.8.2.2.1 Monitoring function:

The following plausibility and activity monitoring is carried out: (e) (7.3.2)

Sensor signal plausibility and signal activity monitoring is performed during coasting conditions during fuel cut-off (slow response / sensor signal range too small).

A malfunction is detected, if the sensor signal is not below a threshold after a short time after entering fuel cut-off. This short time is needed to purge the exhaust pipe. Usually the signal falls from fuel trim correction setpoint (e.g. 680 mV) to a voltage near 0 mV.

If one of the above mentioned malfunctions is detected, the corresponding fault code is stored.

	B1S2	B2S2
Failure during fuel cut-off	P0139	P0159

9.8.2.3 Downstream Oxygen Sensor - Dynamic / Transition Time in Sensor Midpoint Range Monitoring
P1130/1131

9.8.2.3.1 Monitoring Function:

This function monitors the transition Time in sensor midpoint range of the downstream sensor voltage. When a fuel cut-off phase starts, the following steps will be executed:

- sensor voltage must be above a threshold (signal must be rich enough, to measure the swichtime)

Remark: Usually the signal starts at fuel trim control setpoint (e.g. 650 mV)

- sensor voltage value is stored (= "start-value")

- swichtime measurement is started, when the signal is e.g. at 70 % of start value

Remark: The measurement start and stop- value are relative to the start value, to measure always the swichtime around the fuel trim control setpoint.

- swichtime measurement is finished, when the signal is e.g. at 38 % of start value

- measured swichtime ist corrected over mass air flow

The transition time ("swichtime") is represented by a cycle counter. This swichtime is measured over x)* fuel cut-off phases. The mean value after x fuel cut- off phases is compared with a failure threshold.



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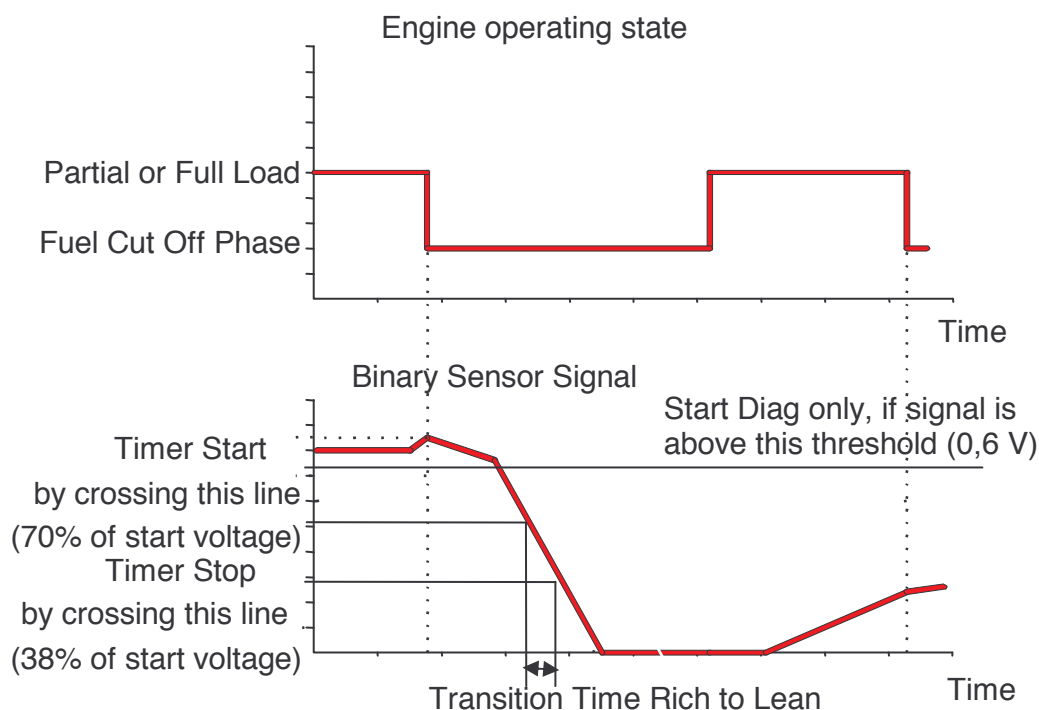
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If this value is above a threshold, a malfunction is detected and the corresponding fault code is stored.

	B1S2	B2S2
Switchtime too high	P1130	P1131

)* For exact values of thresholds etc. please have a look at the summary table!





9.8.2.4 Downstream Oxygen Sensor - Active Signal Check (Stuck lean / rich) *P114A, 114B, 114C, 114D, P2270, 2272, 2271, 2273*

9.8.2.4.1 Monitoring Function:

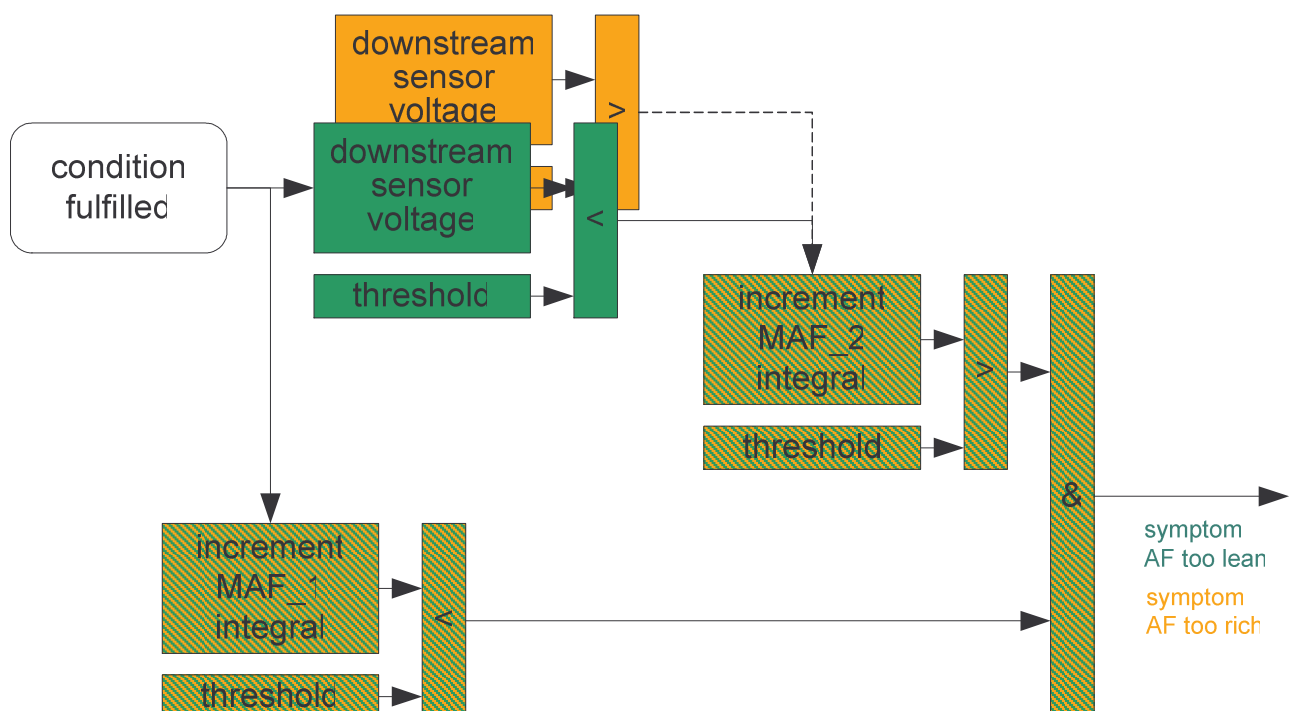
This monitor consists of two parts.

1) Part one monitors the downstream sensor voltage during active fuel trim controller p- share. If the fuel trim control is active, the downstream sensor voltage must be in a band around the trim control setpoint. If the voltage is outside this band)*, a mass air flow integral is incremented. If this integral is over a threshold, a malfunction is detected (see also picture below!). When the downstream sensor voltage will not switch to rich before the integral is over a threshold after a fuel cut-off phase, then this fault will be stored, too.

If the above mentioned malfunctions is detected, the corresponding fault code is stored.

	B1S2	B2S2
Downstream sensor voltage too low	P114B	P114D
Downstream sensor voltage too high	P114A	P114C

)* For exact values of thresholds etc. please have a look at the summary table!





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2) Part two - Downstream Active Test:

This monitor is an enhancement of the Downstream Oxygen Sensor - Active Signal Check (Stuck lean / rich) and the Trim Control Plausibility Monitoring. Its purpose is to help determine the root cause of a stocking downstream sensor signal or a implausible high i- share of the fuel trim controller .

The monitor will only be enabled if a fuel correction fault has been detected and a malfunction code has been stored (P2096 – P2097 – P2098 – P2099)

OR

if the rear sensor signal check has detected, that the rear sensor signal is very rich or very lean and the corresponding malfunction fault code has stored (P114A – P114B – PP114C – P114C)

If one of the listed fault codes exists, this diagnosis will be enabled to determine if the root cause of the malfunction is due to a stuck signal or characteristic line shift of the upstream O2 sensor or due to a stuck signal of the downstream O2 sensor or a system malfunction, i.e. vacuum leak, injector, etc...

If it has been determined that the upstream O2 signal was the root cause of the fuel correction fault, the appropriate DTC will be stored along with the fuel correction or with the downstream sensor stuck DTC (see chapter ' Upstream Oxygen Sensor - Active Signal Check (Shift to lean / rich) ').

If it has been determined that the downstream sensor signal was the root cause of the fuel correction fault, the appropriate DTC (see table below) will be stored along with the fuel correction **or** with the downstream sensor stuck DTC.

This function will also detect, if the oxygen sensor wire harness has been cross connected, i.e., Bank 1 with Bank 2. When this failure is present, the downstream sensor voltages of bank 1 and 2 are on opposite limits.

If one of the above mentioned malfunctions is detected, the corresponding fault code is stored.

	B1S2	B2S2
Downstream sensor stuck rich	P2271	P2273
Downstream sensor stuck lean	P2270	P2272
Downstream sensors interchanged	P0041	



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9.8.2.5 Downstream Oxygen Sensor - Heater Circuit Monitoring

P0141,0161, P0036,0037,0038,0056,0057,0058

9.8.2.5.1 Monitoring Function: Circuit

For proper function of the oxygen sensor, the sensor element must be heated.

A non functioning heater delays the sensor readiness for closed loop control and thus influences emissions.

The monitoring strategy is based on the comparison of the O2 sensor heater resistance to an absolute threshold during coasting conditions where the exhaust temperature is sufficiently low as to cause the sensor ceramic temperature to fall outside normal operating levels, in cases where the heating power is insufficient.

The cooling energy of the exhaust gas is calculated and compared to a calibrated threshold.

The heater monitor is active if the calculated energy is equal or exceeds the threshold.

Then the O2 sensor heater is compared to a calibrated threshold. If the heater resistance is equal or exceeds the threshold, an O2 sensor heater malfunction is detected and the corresponding fault code is stored.

Corresponding fault code:

B1S2 P0141

B2S2 P0161

9.8.2.5.2 Monitoring Function: Power stage

The purpose of this monitor is to detect errors within the O2 Sensor Heater Circuit. The signal for the O2 sensor heater is pulse-width modulated. The signal of the power stage is monitored internally by the driver. The driver can distinguish between three symptoms:

- HO2S Down SCVB
- HO2S Down SCG
- HO2S Down Open Line

If one of the above mentioned symptoms is present, a malfunction is detected and the corresponding fault code is stored.

	B1S2	B2S2
SC Ground	P0037	P0057
SC Vbatt	P0038	P0058
Open circuit	P0036	P0056



9.9 Closed Loop Lambda Control - Enable Conditions

Closed loop lambda control is enabled (with a delay) at the start of a driving cycle and can be temporary or permanently deactivated during the driving cycle. The turn-on delay at the start of a driving cycle is described by the following enable conditions:

- the upstream oxygen sensor operability is detected i.e. the upstream HO2'S operating temperature has been reached
- a calibrated delay time, after end of engine start, has elapsed
- the engine coolant temperature must have exceeded a calibrated threshold or the modeled engine coolant temperature (substitute for a faulty temperature sensor minimum) must have exceeded a calibrated threshold after a calibrated period of time
- no secondary air activity

Closed loop lambda operation is disabled during the driving cycle, if the following operating condition, which permits only mixture enrichment by the lambda controller, is fulfilled:

- when Catalyst overheating prevention is active.

Only for linear Lambda Sensor:

- the A/F ratio set-point value lies below oxygen sensor's measurable limit - in this case only A/F mixture enrichment can be executed by the lambda control (no closed loop operation) the moment the set-point value exceeds the measurable threshold.

Closed loop lambda operation is further deactivated during a driving cycle when any of the following conditions are fulfilled:

- during fuel cut-off or cylinder shut-off and immediately afterwards till the oxygen sensor again starts indicating correct values (the waiting time depends on integrated mass airflow or on a calibrated delay time).
- the mass air flow is below a calibratable threshold that leads to the minimum possible injection time.
- Secondary air system active



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9.10 Engine Coolant Temperature (ECT) Monitoring

1968.2 Sec (e) (10.1)

9.10.1 Electrical Coolant Temperature Diagnosis
P0117,0118

9.10.1.1 General Description

The purpose of this diagnosis is to detect electrical faults of the sensor signal. The input signal is analog from a NTC and has to be in a calibratable range. Short cut to ground can be detected immediately, short cut to voltage battery or open load after a delay time. If an error symptom is detected, the error counter is de-bounced.

9.10.2 Coolant Temperature Gradient Diagnosis
P3198

9.10.2.1 General Description

The purpose of this diagnosis is to detect an implausible gradient on the coolant temperature signal. The diagnostic function checks whether the difference between one measured coolant temperature value and the succeeding value is too big.

9.10.3 Coolant Temperature Stuck Diagnosis
P3199

9.10.3.1 General Description

The purpose of this diagnosis is to detect a stuck coolant temperature signal. The diagnostic function checks if after a variation of the calculated coolant temperature also a variation of the measured coolant temperature is detected.

For RBM handling the Cold Start Denominator will be considered.



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**9.10.4 Coolant Temperature Sensor stuck in a range Diagnosis with Engine off timer
*P316A***

9.10.4.1 General description

The engine off time is calculated by a relative time counter of an instrumentation CAN message. After evaluation of this message the engine off time is compared with the difference of ECT at engine stop and ECT. This is to check if the engine off time is within a certain temperature range.

**9.10.5 Engine off timer Monitoring
*P1515***

9.10.5.1 General description

The engine off time is calculated by a relative time counter of an instrument CAN message. After evaluation of this message the engine off time is compared with the difference of ECT at engine stop and ECT. This is to check if the engine off time is within a certain temperature range.



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9.11 Thermostat

P0128,0597,0598,0599

9.11.1 Description of the Engine Coolant Thermostat Monitoring

The coolant thermostat monitoring is done to detect a slow warm-up due to heat losses through thermostat and radiator. It is based on the comparison of the measured ECT sensor signal and the calculated ECT model (TCO_SUB).

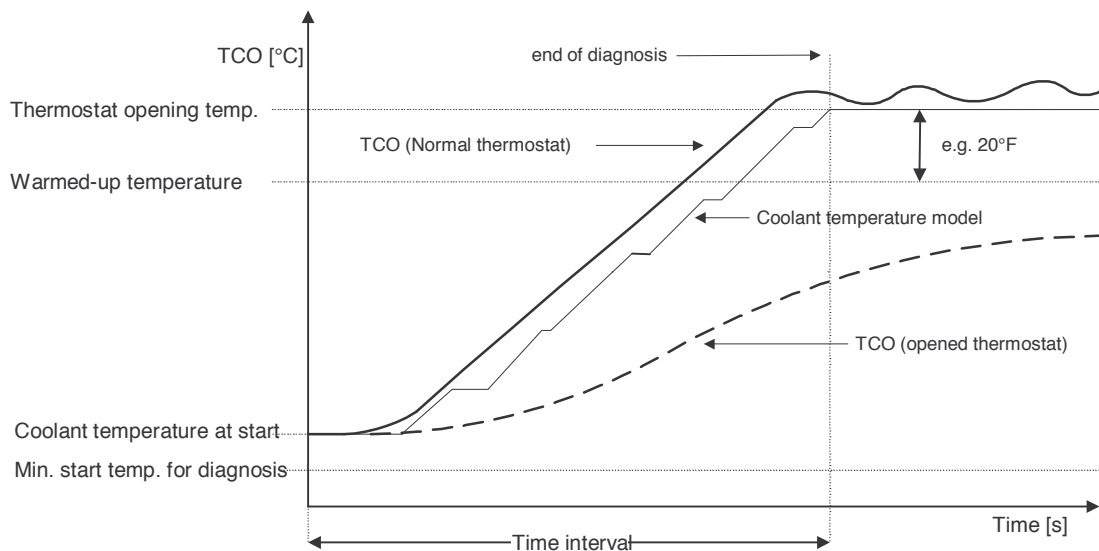
The ECT model calculation is depending on the speed of the waterpump, engine load and ambient temperature.

A malfunctioning coolant thermostat is detected, if the calculated ECT model has exceeded the threshold 1 (P0128) and the measured ECT sensor signal remains below threshold 2 (P0128).

Before a malfunctioning coolant thermostat is entered into failure memory, the conditions concerning low load, coasting duration and IAT during the monitoring are checked. If the monitoring conditions are met, the coolant thermostat is entered into failure memory. Otherwise the coolant thermostat monitoring is inhibited for this driving cycle.



9.11.2 Example of Monitoring Method:



A comparison between the measured coolant temperature and the “warmed-up temperature” is done after a specific time interval. The interval itself is based on the coolant temperature model (Function of intake air mass).

As soon as the model temperature exceeds the thermostat opening temperature and all other monitoring conditions are fulfilled at the same time, a valid diagnosis occurs.

At that time, if the measured coolant temperature is higher than warmed-up temperature, the thermostat is concluded as normal thermostat.

On the contrary, if the measured coolant temperature is lower than warmed-up temperature, the thermostat is concluded as opened stuck thermostat.



9.12 Intake Air Temperature (IAT) Monitoring

9.12.1 Electrical Intake Air Temperature Diagnosis *P0112,0113*

9.12.1.1 General Description

The purpose of this diagnosis is to detect electrical faults as defined in OBDI requirements. The input signal is analog from a NTC and has to be in a calibratable range. Short cut to ground can be detected immediately, short cut to voltage battery or open load after a delay time. If an error symptom is detected, the error counter is de-bounced.

9.12.2 Intake Air Plausibility Check *P0111,111E,111F*

9.12.2.1 General Description

This diagnosis checks IAT integrity for a plausible range and / or signal stuck. For the range detection, IAT has to be within coolant temperature and ambient temperature window. If IAT is outside of the range plus an offset, the error symptom is set and the error counter is de-bounced.

If the vehicle was driven with a certain vehicle speed for a calibratable time (cool down of hot IAT) and afterwards the vehicle was in idle for a calibratable time, and the IAT signal has not moved, then a stuck IAT signal is detected and the error is de-bounced.

For RBM handling the Cold Start Denominator will be considered.



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9.13 Camshaft and Crankshaft Sensor including Valve Timing

9.13.1 Camshaft Sensor

**9.13.1.1 Variable Camshaft Timing (Vanos) (detection of mechanical IVVT error)
*P0012 /0015***

The BMW-Vanos is a combined hydraulic and mechanical camshaft control unit, managed by the ECU. The double Vanos allows the engine to control valve-timing continuously for both intake and exhaust camshafts. The electronically control of the Vanos positions is dependant on engine speed, load and temperature.

The diagnosis is monitoring the correct mechanical function of the variable camshaft timing. The diagnosis carries out a continuous rationality check of the Vanos function. If a malfunction is detected, an error bit will be set and sent to the Error management module. This module produces the final information for setting the corresponding DTC.

The diagnostic strategy for inlet and exhaust camshaft is identical.

9.13.1.1.1 Control deviation of the camshaft position controller: ("target + slow response")

9.13.1.1.2 Description:

In this diagnosis module the difference between the actual and target position of the Vanos units ("control deviation") is checked. If the calculated difference between these two positions exceeds the established threshold, a counter is started. The counter is incremented twice per crank revolution (but not exceeding 10 msec-rates). If the counter exceeds a limit (also adjustable), a Rationality Fault (DTC) is stored.

The control deviation diagnosis has got an interface to the Rate-Based Monitoring module.

A. In-use monitor performance Ratio:

The incrementing of the numerator, denominator, and the ratio calculation for the Variable Camshaft Timing monitor is executed by the Rate-Based Monitoring module. Like all monitors for which a standardized track and report in-use performance is required, the Variable Camshaft Timing monitor reports to the RBM-module via status flags - see description of RBM module.

B. Conditions for incrementing the Numerator:

The numerator is incremented if and only if the monitor is not inhibited due to stored faults and the diagnostic has been performed and a fault would have been detected.



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C. Conditions for incrementing the Denominator:

The denominator is incremented if the monitor is not inhibited due to stored faults, the general driving conditions have been fulfilled and all additional physical conditions for incrementing have been fulfilled.

9.13.1.2 Camshaft position sensor (CMP)

P0340/0365/1300/130A/0344/0369/1554/1553/0016/0017

9.13.1.2.1 Description:

The purpose of the diagnosis is to detect when the camshaft reference position is outside the designed range relative to the engine position from crankshaft and to detect a signal which is not valid.

The diagnostic strategy for inlet and exhaust camshaft is identical.

9.13.1.2.2 Monitoring Function

The detection of each camshaft position is done by an active hall sensor and a cam wheel, "3 asymmetric teeth". The camshaft sensor delivers 3 high and 3 low phases of different length per 720°CRK. The high or low pegel of the signal at the reference gap of the crankshaft signal determines the position of the engine within the combustion cycle. With that information, a engine position is calculated from the crankshaft position sensor within a range from [0 to 720[°CRK

The following malfunctions are detected:

- CMP sensor signal plausibility	P0340 / P0365
- CMP sensor signal segment period	P1300 / P130A
- CMP sensor signal loss of synchronization	P0344 / P0369
- CMP sensor signal reference to CRK position	P1554 / P1553
- CMP sensor signal jump of chain	P0016 / P0017



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9.13.1.2.3 Diagnosis of signal plausibility

P0340/0365

The monitor checks once per combustion cycle the edge counter of the camshaft. If the edge counter has not changed during the last cycle, a cycle counter is incremented. When the counter reaches a threshold, the error CAM_plaus is delivered to the error management.



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9.13.1.2.4 Diagnosis of period length

P1300/130A

The monitor checks at every edge of the CMP signal the length of the last signal period. If the difference to the designed length exceeds a max value, the corresponding debounce counter is incremented. When the counter reaches a threshold the error CAM_period is delivered to the error management.



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9.13.1.2.5 Diagnosis of synchronization state

P0344/P0369

The monitor checks at every reference gap of the crankshaft sensor signal whether the camshaft is synchronized or not. If the camshaft is not synchronized, a revolution counter for the camshaft is incremented. When the counter reaches a threshold the error CAM sync is delivered to the error management.



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9.13.1.2.6 Diagnosis of mechanical reference position

P1554/1553

The monitor checks at least once per driving cycle the position of the camshaft in reference position compared to the crankshaft position. The deviation of all camshaft edges compared to the designed position is averaged. If the average value exceeds a max value, the corresponding debounce counter is incremented. When the counter reaches a threshold the error CAM_ref_crk_cam is delivered to the error management.



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9.13.1.2.7 Diagnosis of mechanical chain jump

P0016/0017

The diagnosis is performed after the reference position adaptation and uses the same calculated average value. The average value is stored in the non volatile RAM of the ECU as an adaptation value. Before storing the value, the new adapted value is compared with the stored value. If the deviation exceeds a max value, the error CAM_one_tooth_off is delivered to the error management and the new value is not stored in RAM. With this diagnosis a chain jump of the timing chain is detected.

The diagnosis of the mechanical chain jump has got an interface to the Rate-Based Monitoring module.

A. In-use monitor performance Ratio:

The incrementing of the numerator and denominator for the diagnosis of chain jump is executed by the Rate-Based Monitoring module. Like all monitors for which a standardized track and report in-use performance is required, the diagnosos monitor reports to the RBM-module via status flags - see description of RBM module.

B. Conditions for incrementing the Numerator:

The numerator is incremented if and only if the monitor is not inhibited due to stored faults and the diagnostic has been performed and a fault would have been detected.

C. Conditions for incrementing the Denominator:

The denominator is incremented with every driving cycle.

Flowchart for diagnosis of chain jump:
see 9.14.1.2.7 Diagnosis of mechanical reference position



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9.13.1.3 Camshaft Crankshaft synchronization

P0341

9.13.1.3.1 Description:

The purpose of the diagnosis is to determine whether the inlet (first) camshaft is used for engine synchronization or the exhaust camshaft.

9.13.1.3.2 Monitoring Function

The diagnosis is performed at every edge of the selected camshaft signal and at the reference gap of the CKP sensor signal. With the period between current event and the last event, the function eliminates the edges of the camshaft which are not inside the designed window. The window is calculated from the designed position plus / minus a tolerance. The tolerance is expanded by the range of the variable valve timing, when the camshaft is not in lock position.

The following malfunctions are detected:

- CMP sensor signal not valid for synchronization

P0341



9.13.1.3.3 Diagnosis of camshaft crankshaft synchronization

The monitor eliminates with every event the edges from the list of all 6 cam edges, which are not inside the pattern. If only one edge is left, the edge number is identified and the camshaft is valid for synchronization. If no edge is left in the list, synchronization failed and is started again. If a calibrated number of synchronizations failed, the error is delivered to the error management. Afterwards, an only if synchronization fails with the intake camshaft, the same procedure is started with the exhaust camshaft.

9.13.2 Crankshaft position sensor (CRK) *P0335/0370/0373*

9.13.2.1 Description:

Sec (e) (13) & (e) (16.2.1) (B – C)

The purpose of this diagnostic is to check the integrity of the crankshaft sensor signal and/or electrical malfunctions. (Open line, SCG, SCVB)

9.13.2.2 Monitoring Function

The detection of crankshaft position is done by an active hall sensor and a crank wheel, "e.g. 60 minus 2 teeth". A reference gap, "e.g. of two teeth" allows the detection of the top dead center of cylinder 0. The crankshaft sensor delivers a certain number of high and low phases per 360°CRK. The transition from high to low is a falling edge; from low to high is a rising edge. Only the falling edges are counted. The difference between two falling edges is 6° CRK.

The following malfunctions are detected:

- missing CRK sensor signal P0335
- no plausible CRK signal P0335
- wrong tooth number P0370
- wrong tooth period P0370
- sync error P0373

A teeth counter is incremented at every falling edge of the CRK sensor signal. If plus or minus one tooth is detected during the last 360° CRK at the reference gap, the tooth number debounce counter will be incremented. If the counter exceeds a limit, a CRK tooth error is delivered to the error management.

If more than one tooth plus or minus is detected the CRK loses synchronization and a CRK sync debounce counter will be incremented. If the counter exceeds a limit, a CRK sync error is delivered to the error management.

The detection of a tooth period error is done by an acceptance window. The expected tooth period is multiplied and divided with an engine speed dependency factor. The result is a bottom and a top limit of tooth period, in which the transition from high to low



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of the electrical signal has to occur. If a tooth period is not valid, the tooth period error debounce counter will be incremented. If the counter exceeds a limit, a CRK tooth per error is delivered to the error management.

Detection of implausible crankshaft signal is based on the detection of CAM signals without receiving correct CRK signal. If 12 or more CAM edges are detected (eg. 2 working cycles), without valid synchronization of the crankshaft, then CRK plaus error is detected and delivered to the error management. If no CRK signal at all is received, the symptom is "missing signal", else the symptom is "implausible signal".



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9.14 Variable Valve Lift - Electrical Diagnosis

P1047,1048,1049,1076,1075,103A,1078,107A,107B107C,105B,105A,1055,1056,1019,1020,1057,1017,1030,1064,101A,1023,1024,1041

9.14.1 Descriptions:

The electronic control of the Variable Valve Lift positions is dependant on, Voltage Limits, Start & Stop position, Adaptations, current and temperature. The following errors will be detected in this system.

Electrical diagnosis is performed internally to the controller in order to detect the following errors: Short circuit to battery, to ground or short circuit to each other:

SCVB – P1047
SCG – P1048
S together – P1049

System current is monitored through the ECU and the variable valve train relay. This diagnosis checks for over-temperature and overload conditions and performs short term and long term high current monitoring (looks for current spikes over a threshold, and time vs. current table based):

Relay over temp – P1076
Relay overload – P1075
Relay overload – P103A
VVL Bus Conductor Temp – P1078
VVL Power Stage Warning – P107A
VV Train Bus Conductor Temp – P107B
VVL Power Stage Warning – P107C

Powerstage self diagnosis is performed to realize if power stage diagnosis has detected under voltage or overcurrent of high or lowside. The diagnosis is an self check realized therefore in the specific hardware (powerstage).

undervoltage of driver – P105B
overcurrent detection on high side / low side of H-bridge – P105A

Power Supply Control Motor is monitored through the main relay and checks over and under voltage conditions. There is also a Power Supply sensor diagnosis which checks short circuit to battery and to ground. If this occurs, the following DTC's will be stored:

Power Supply over voltage – P1055
Power Supply Under Voltage – P1056
Power Supply Sensor SCVB – P1019
Power Supply Sensor SCG – P1020

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VVL Relay Diagnosis is performed internally to the ECU and does a comparison of the main relay voltage to the variable valve lift capacitors. If the difference is greater than a threshold then an appropriate DTC will be stored.

Relay Diagnosis – P1057 (Battery Main Relay Voltage – Capacitor Voltage)

Sensor Diagnosis is performed internally to the ECU and checks the sensor supply voltage to tunable boundaries. A sensor signal versus an internal sensor self check will determine the integrity of the sensor. If an error is detected, the appropriate DTC will be stored.

Sensor Signal fault – P1017 (ECU check)

Control Position Diagnosis and the **Value Comparison Diagnosis** are used to monitor the start and stop positions and the current PWM signal movement of the variable valve train system. If the difference between the target and actual angle is determined to be greater than a threshold percentage an error is detected, if a difference in the start and stop positions are greater than a threshold an error is detected. If any of these errors is detected, the appropriate DTC will be stored:

Control Position fault – P1030 (PWM out of range)

Value Comparison fault – P1064 (Difference)

Self Learning / Adaptation Diagnosis is performed internally to the ECU. There are three adaptation diagnoses performed in this function, the top and bottom limit individually out of range and both limits out of range. Furthermore the ECU self check diagnosis is performed, which is basically a check sum error. If any of these errors is detected, the appropriate DTC will be stored:

Top Limit Fault – P101A (top limit not reached)

Bottom Limit Fault – P1023 (bottom limit not reached)

Both Adaptations Fails – P1024 (both limits not reached)

ECU Check Sum Error – P1041



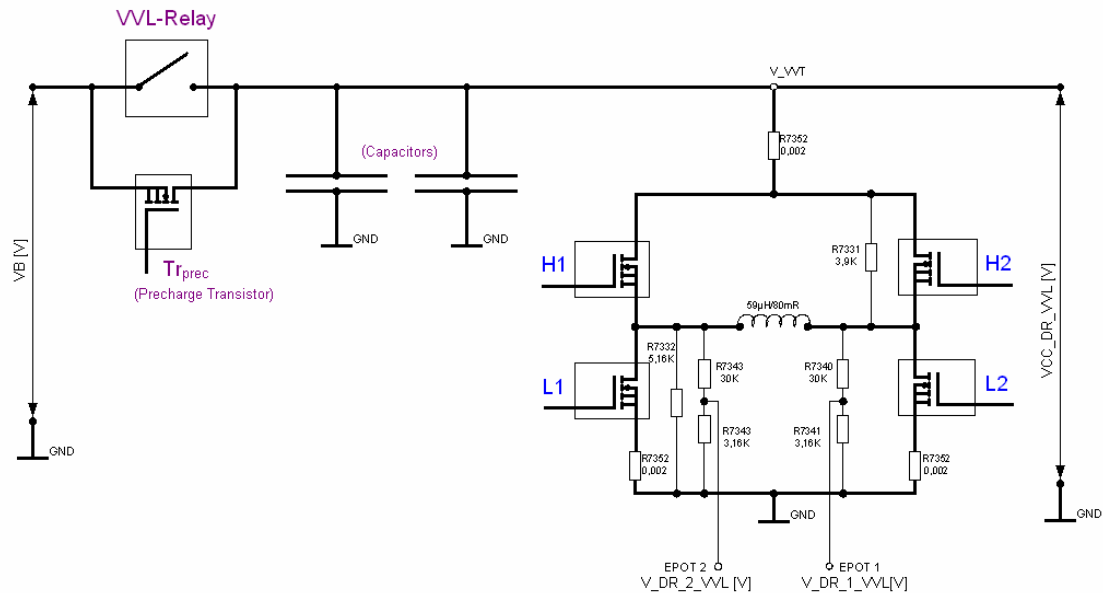
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9.14.2 Variable Valve Lift Electrical Diagram:



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9.15 Electronic Throttle Control Monitor

9.15.1 Electronic Throttle Control (ETC) Motor Control Circuit

9.15.1.1 Monitoring Descriptions:

P1636

ETC Driver diagnosis (H-bridge): The ETC - H-Bridge IC continually checks the MTC if there is a short circuit to battery voltage or ground. In addition the IC is able to detect overtemperature. This is performed internally to the ECU.

9.15.1.2 ETC spring check (start routine):

(P169A, P1694)

This Diagnosis checks if the throttle spring is working correctly and if the throttle limp home position can be reached.

The diagnosis is performed at the beginning of every driving cycle at ignition "Key ON" position.

9.15.1.3 ETC adaptation diagnosis:

(P1632, P1633, P1634, P1635, P1644)

After the initial engine start and / or component change, the characteristic Potentiometer values for the limp home position and the lower mechanical stop are learned within an adaptation routine. The values are stored at the end of the driving cycle in the non-volatile memory.

If the conditions are not fulfilled, the malfunction errors (DTC's) are stored.

9.15.1.4 Electronic Throttle Control (ETC) Motor Control Performance

(P1637, P1639)

This diagnosis is able to detect a too slow or jammed actuator. The given pulse width modulation signal (MTCPWM) exceeds the position controller permissible maximum value for longer than designated (Max short or Max Long) time.

If either of the times is exceeded, the appropriate DTC will be stored.

Also if a maximum allowed difference between throttle actual value and setpoint value is exceeded, a DTC is stored.



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9.15.1.5 Electronic Throttle Control (ETC) air supply rationality check
P1417

If P1639 active OR
P1637 active OR
P1636 active OR
P1632 active OR
P1633 active OR
P1694 active OR
P1644 active OR
P1634 active OR
P169A active OR
P1635 active OR
[(P0122 active OR P0123 active) AND (P0222 active OR P0223 active)]

the composite error P1417 will be stored.



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9.16 ISC (Idle Speed Control) Actuator

P0506,0507

9.16.1 Idle Speed Control Electrical Diagnosis

Sec (e) (16.2.2) (B) CCR Title 13, 1968.2

9.16.1.1 General Description:

The purpose of this diagnosis is to detect electrical faults of the idle speed controller circuit. The controller is PWM modulated, inverse output signals are used for continuous activation of the opening and closing of a double-wound coil actuator.

The error detection is activated at Key "ON". The following errors are detected:

- Short to ground
- Short to battery
- Open circuit

If any of the above errors are detected during opening or closing of the controller, the appropriate DTC will be stored.

9.16.1.2 Idle Speed Control - Rationality Diagnosis

General Description:

This diagnosis detects the stability of the idle speed. If the commanded idle speed and the actual idle speed is not within a calibratable value, above or below commanded, the idle speed is considered to be out of range.

If the above failure conditions are detected, the appropriate DTC will be stored.



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9.17 Air Mass Flow Meter - Rationality check

P1415, 1424

ARB CFR Title 13, 1968.2 Sec. (e) (16.2.1)

9.17.1 Description

Depending on engine speed, valve-lift, inlet camshaft position, outlet camshaft position and manifold pressure an air mass flow into the cylinder is calculated. There is also a correction of the calculated air mass flow depending on intake air temperature, coolant temperature and ambient pressure. The ratio between the measured air mass flow and the calculated air mass flow must be between calibratable MIN/MAX-values. If the MIN/MAX thresholds are exceeded, a time counter is incremented. After this counter reaches the threshold within one diagnosis cycle, an air mass flow meter malfunction is detected.

9.18 Manifold Differential Pressure Sensor - Rationality check

P1124, P1104, 1105

ARB CFR Title 13, 1968.2 Sec (e) (16.2.1)

9.18.1 Description

For a variable valve lift engine, the main function of the throttle body is to control the pressure in the intake manifold. Therefore the manifold differential pressure plausibility check is testing the plausibility of measured intake manifold pressure in comparison to the measured throttle position. So no throttle position acquisition error must be present.

The setpoint of the differential intake manifold pressure is up to 60-70% of maximum torque request constant 5 kPa beneath ambient pressure. At higher load the differential pressure setpoint becomes Zero.

In case the setpoint of the differential pressure in the intake manifold is > 3 Kpa beneath the ambient pressure, a differential pressure controller is active. The output of the controller is monitored. The output of the manifold pressure controller has to be between calibratable MIN/MAX-thresholds. If the MIN/MAX thresholds are exceeded a time counter is incremented. After this counter reaches the threshold within one diagnosis cycle, a manifold differential pressure sensor malfunction is detected.



9.19 Comprehensive Component Monitoring OBD II Siemens VDO function definition

9.19.1 Strategy

Principle:

- o **Sensors** that can affect emissions or are used to monitor other component / system are monitored for circuit continuity and short to battery voltage and / or to ground using high and low voltage signal limit.
- o **Actuators** that can affect emissions or are used to monitor other component / system are monitored by power stage voltage check for valid signals.
- o For some of sensors or actuators, plausibility checks are included to ensure proper operation of the components.

9.19.2 Monitoring Conditions for electrical check:

- o Continuous Monitoring

9.19.3 Monitoring Strategy for sensors:

- o Sensor signals out of a defined range are regarded as circuit malfunctions shorted to BATT, GND or Open circuit.

9.19.4 Monitoring Strategy for actuators:

- o Invalid actuator output signals at power stage are regarded as circuit malfunctions shorted to BATT, GND or Open circuit.

9.19.5 Rationality Check:

- o Components are checked for the integrity of their values. This is accomplished by the use of a model or other sensor inputs. If a component does not function as expected or the integrity is in question (values are not within a threshold) it is considered out of range / plausible.



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- 9.19.6 Throttle Position Sensor - Electrical Check**
P0122, 0123, 0222, 0223
- 9.19.7 Throttle Position Sensor - Plausibility Check**
P110D
- 9.19.8 Mass Air Flow Sensor - Electrical Check**
P116C, 116E
- 9.19.9 Camshaft Position Actuator – Electrical Check**
P0010, 0013, 2088, 2090, 2089, 2091
- 9.19.10 Vehicle Speed Signal**
P0500, 0503
- 9.19.11 Injector Valve Diagnosis**
P0201, P0202, P0203, P0204, P0205, P0206, P0261, P0262, P0264, P0265, P0267, P0268, P0270, P0271, P0273, P0274, P0276, P0277
- 9.19.12 Differential Pressure Sensor - Electrical check**
P1197, 1198
- 9.19.13 Knock Sensor diagnosis**
P0326, 0327, 0328, 1327, 1328, 135B
- 9.19.14 CAN Communication**
U112A, 112B, 1101, 110E, 110F, 1110
- 9.19.15 Pedal Position Sensor**
P2120, 164C, 1625
- 9.19.16 Ambient Pressure Sensor - Electrical check**
P2228, 2229
- 9.19.17 Ambient Pressure Sensor - Rationality Check**
P321E, 321F
- 9.19.18 EVAP – DMTL Valve, Electrical check**
P2418, 2419, 2420
- 9.19.19 EVAP – DMTL Heater, Electrical check**
P240A, 240B, 240C
- 9.19.20 EVAP – DMTL Pump, Electrical check**
P2400, 2401, 2402
- 9.19.21 EVAP – Purge Control Valve, Electrical check**
P0444, 0458, 0459



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9.20 Listing of all ECM Input and Output Signals

BMW signal naming	BMW N52KP	Pin	SIEMENS signal naming	MSV80	OBD II relevant
Fahrzeug CAN-Schnittstelle 1 LOW	D_PT_CANL1	1_01	CAN-Low1	CAN1_L	No
Start(er)-Relais (Automatikstart)	A_S_START	1_02	start relay	RLY_START	No
Generatorschnittstelle	D_BSD	1_03	generator interface	BSD	No
Bremslichtschalter	E_S_BLS	1_04	brakelight switch	BLS	No
Abgasklappe	A_S_AKL	1_05	exhaust flap	EF	No
Masse Temperatur Kühlwasseraustritt	M_TKA	1_06	ground coolant outlet temperature	TCO_EX_GND GND	Yes
Fahrerwunsch Geber 2	E_A_FWG2	1_07	pedal value sensor 2	PVS_2	No
Elektr. Lüfter getaktet	A_T_ELUE	1_08	cooling fan	CFA	No
Luftklappe	A_T_LKS	1_09	air flap	AF	No
Masse Pedalwertgeber 1	M_FWG1	1_10	ground pedal value sensor 1	GND	No
Spannungsversorgung 5V (PWG1)	A_U_FWG1	1_11	supply voltage PVS1	PVS1_VCC	No
Lin Bus	LIN_BUS_MS	1_12	Lin Bus	LIN	No
Sekundärluftpumpe Stufe 1	A_S_SLP	1_13	Secondary air pump	SAP	not used
Fahrzeug CAN-Schnittstelle 1 HIGH	D_PT_CANH1	1_14	CAN-High1	CAN1_H	No
Wegfahrsperr, EWS4	D_EWS	1_15	imobilizer EWS4	IMOB	No
Bremslichtschalter	E_S_BLTS	1_16	brakelight test switch	BTS	No
Fahrzeuggeschwindigkeit	E_F_DFAHR	1_17	wheel speed	WHEEL	Yes
Kupplungsschalter	E_S_KUP	1_18	clutch switch	CLU_SWI	No
Temperaturfühler Kühlwasseraustritt	E_A_TKA	1_19	coolant outlet temperature	TCO_EX	not used
Fahrerwunsch Geber 1	E_A_FWG1	1_20	pedal value sensor 1	PVS_1	No
Drehzahl	A_F_TD	1_21	engine speed signal	ESS	Yes
Fzg. Pin Kl 15/3	E_S_KL15_3	1_22	Ignition key Kl. 15/3	V_IG_3	No
Masse Pedalwertgeber 2	M_FWG2	1_23	ground pedal value sensor 2	GND	No
Spannungsversorgung 5V (PWG2)	A_U_FWG2	1_24	supply voltage PVS2	PVS2_VCC	No
Sekundärluft	E_A_HFMS	1_25	mass air flow metersecondary air	MAFMS	not used
EBox-Lüfter	A_S_EBOXL	1_26	cooling fan Ebox	CFA_EBOX	No
Fzg. Pin Kl.15	E_S_KL15	2_01	ignition key Kl.15	V_IG	No
Lin Bus	LIN_BUS_MS	2_02	Lin Bus	LIN	No
Fahrdynamikkontrolle	E_A_FDC	2_03	Sound flap switch	SOF_SWI	No
Multifunktionslenkrad	D_FGRD	2_04	multifunctional steering wheel	MSW	No
Pumpstrom, Stetige-Lambdas. v Kat 2	A_I_LSVP2	2_05	pump current output 2	LSL_IA_2	Yes
Pumpzelle, Stetige-Lambdas. v Kat 1	E_A_LSVP1	2_06	pump current measurement 1	LSL_IP_1	Yes
Pumpzelle, Stetige-Lambdas. v Kat 2	E_A_LSVP2	2_07	pump current measurement 2	LSL_IP_2	Yes
Lambdasonde/ Referenzzelle vor Kat 1	E_A_LSVR1	2_08	lambda sensor upstream 1	LS_UP_1	Yes
Lambdasonde/ Referenzzelle vor Kat 2	E_A_LSVR2	2_09	lambda sensor upstream 2	LS_UP_2	Yes
Masse Lambdasonde vor Kat 1	M_LSV1	2_10	ground lambda sensor upstream 1	LS_UP_1_GND	Yes
Masse Lambdasonde vor Kat 2	M_LSV2	2_11	ground lambda sensor upstream 2	LS_UP_2_GND	Yes
Heizung Lambdasonde vor Kat 1	A_T_LHV1	2_12	lambda sensor heater upstream 1	LSH_UP_1	Yes
Heizung Lambdasonde vor Kat 2	A_T_LHV2	2_13	lambda sensor heater upstream 2	LSH_UP_2	Yes
Haupt-Relais (ÄAnsteuerung)	A_S_HR	2_14	Main relay	RLY_MAIN	No
Ventil Tankleckdiagnose	A_S_DMTLV	2_15	Tank leakage detection valve	DMTLV	Yes
Pumpe Tankleckdiagnose	A_S_DMTLP	2_16	Tank leakage detection pump	DMTLP	Yes
DMTL Heizung	A_S_DMTLH	2_17	DMTL heater	DMTLH	Yes
Pumpstrom, Stetige-Lambdas. v Kat 1	A_I_LSVP1	2_18	pump current output 1	LSL_IA_1	Yes
Lambdasonde hinter Kat 2	E_A_LSH2	2_19	lambda sensor downstream 2	LS_DOWN_2	Yes
Lambdasonde hinter Kat 1	E_A_LSH1	2_20	lambda sensor downstream 1	LS_DOWN_1	Yes
Relais Klimakompressor	A_S_KOREL	2_21	relay air conditioning compressor	RLY_ACC	No
Reserve Analogeingang 1	E_A_RES1	2_22	Reserve analog 1	SPARE_AN_1	Not used
Masse Lambdasonde hinter Kat 1	M_LSH1	2_23	ground lambda sensor downstream 1	LS_DOWN_1_ GND	Yes
Masse Lambdasonde hinter Kat 2	M_LSH2	2_24	ground lambda sensor downstream 2	LS_DOWN_2_ GND	Yes
Heizung Lambdasonde hinter Kat 2	A_T_LHH2	2_25	lambda sensor heater downstream 2	LSH_DOWN_2	Yes

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Heizung Lambdasonde hinter Kat 1	A_T_LHH1	2_26	lambda sensor heater downstream 1	LSH_DOWN_1	Yes
Dauerplus Kl.30	E_U_30	3_01	direct battery Kl.30	VB	No
Hauptrelais	E_U_HR	3_02	main relay Kl.87	V_EL	No
Masse Zündung	M_ZUE	3_03	Ground ignition	GND_IG	No
Masse Elektronik Einspritzventile	M_EL/EV	3_04	ground electronic, injection	GND_EL	Yes
Masse VVT	M_VVT	3_05	ground VVT	GND_VVT	Yes
Masse VVT	M_VVT	3_06	ground VVT	GND_VVT	Yes
Spannungsversorgung VVT	E_U_VVTR1	4_01	Supply voltage from VVT relay	V_VVT	Yes
Spannungsversorgung VVT	E_U_VVTR1	4_02	Supply voltage from VVT relay	V_VVT	Yes
Motorausgang 2 VVT	A_T_VVT2M1	4_03	Motor output 2 VVT	VVT2M1	Yes
Motorausgang 1 VVT	A_T_VVT1M1	4_04	Motor output 1 VVT	VVT1M1	Yes
Motorausgang 2 VVT	A_T_VVT2M1	4_05	Motor output 2 VVT	VVT2M1	Yes
Motorausgang 1 VVT	A_T_VVT1M1	4_06	Motor output 1 VVT	VVT1M1	Yes
Masse (nicht angeschlossen)	n. c. (Masse)	5_01	GND (not connected)	n. c.	not used
Masse (nicht angeschlossen)	n. c. (Masse)	5_02	GND (not connected)	n. c.	not used
Masse (nicht angeschlossen)	n. c. (Masse)	5_03	GND (not connected)	n. c.	not used
MAF Frequenzsignal	E_P_HFM	5_04	SIMAF	SIMAF	not used
Masse Öldrucksensor	M_OLD	5_05	Ground oil pressure sensor	OILP_GND	not used
Reserve Analogeingang 3	E_A_RES3	5_06	Reserve analog 3	SPARE_AN3	not used
Spannungsversorgung 5V (Öldrucksensor)	A_U_OLD	5_07	Supply voltage OILP	OILP_VCC	No
NTC-Wasser (Motortemperatur)	E_A_TMOT	5_08	Coolant temperature	TCO	Yes
Masse Motortemperaturfühler	M_TMOT	5_09	ground coolant temperature sensor	TCO_GND	Yes
Öldruck	E_S_OLD	5_10	Oil pressure	POIL	No
Öldruckventil	A_T_OLP	5_11	Oilpressure valve	SAV_OILP	not used
Kraftstoffpumpe	A_S_EKP	5_12	Electrical fuel pump	EFP	No
Haupt-Relais (Ansteuerung)	A_S_HR	5_13	Main relay	RLY_MAIN	No
Spannungsversorgung 5V (DKG1,2)	A_U_DKG	5_14	supply voltage TPS	PVS1TPS_VC C	Yes
Ansteuerung 1 Drosselklappe	A_T_MDK1	5_15	throttle actuator out 1	MTC1	Yes
Ansteuerung 2 Drosselklappe	A_T_MDK2	5_16	throttle actuator out 2	MTC2	Yes
Masse reserve 2	M_RES1	5_17	Ground spare 2	SPARE2_GND	not used
Schaltsgaugrohr 2	A_T_DISA2	5_18	variable intake manifold 2	VIM2	No
Klopfsensor 1B (Diff.- Signal)	E_A_KS1B	5_19	knock sensor 1B	KNKS_1_B	Yes
Klopfsensor 2B (Diff.- Signal)	E_A_KS2B	5_20	knock sensor 2B	KNKS_2_B	Yes
Applikation CAN-Schnittstelle 3 HIGH	D_APPLI_CANH	5_21	CAN-High3	CAN3_H	No
Lokaler CAN-High	D_LO_CANH	5_22	local CAN-High	LOCAN_H	No
Tankentlüftungsventil	A_T_TEV	5_23	canister purge solenoid	CPS	Yes
Soundklappe	A_S_ESK	5_24	sound flap	SF	No
Spannungsversorgung 5V (Reserve)	A_U_RES1	5_25	Supply voltage spare	SPARE_VCC	not used
Reserve Analogeingang 2	E_A_RES2	5_26	Reserve analog 2	SPARE_AN_2	not used
Masse Heißluftmassenmesser	M_HFM	5_27	ground mass air flow meter	MAFM_GND	Yes
Ansauglufttemperatur	E_A_TANS	5_28	intake air temperature	TIA	Yes
Kurbelwellensensor	E_P_KWG	5_29	crankshaft position sensor	CRK	Yes
Masse Kurbelwellensensor	M_KWG	5_30	ground crankshaft position sensor	CRK_GND	Yes
Spannungsversorgung 5V (SDF)	A_U_SDF	5_31	supply voltage MAP	MAP_VCC	Yes
Masse Saugrohrdrucksensor	M_SDF	5_32	ground manifold air pressure	MAP_GND	Yes
Saugrohrdrucksensor	E_A_SDF	5_33	manifold air pressure	MAP (IAP)	Yes
Reserve Analogeingang 1	E_A_RES1	5_34	Reserve analog 1	SPARE_AN_1	not used
Generatorschnittstelle	D_BSD	5_35	generator interface	BSD	No
Drosselklappengeber2	E_A_DKG2	5_36	throttle position sensor 2	TPS_2	Yes
Drosselklappengeber1	E_A_DKG1	5_37	throttle position sensor 1	TPS_1	Yes
Masse Drosselklappengeber	M_DKG	5_38	ground throttle position sensor	TPS_GND	Yes
Öldrucksensor	E_A_OLD	5_39	Oil pressure sensor	OILP	not used
Schaltsgaugrohr1	A_T_DISA1	5_40	variable intake manifold 1	VIM1	No
Klopfsensor 1A (Diff.- Signal)	E_A_KS1A	5_41	knock sensor 1A	KNKS_1_A	Yes
Klopfsensor 2A (Diff.- Signal)	E_A_KS2A	5_42	knock sensor 2A	KNKS_2_A	Yes
Applikation CAN Schnittstelle 3 LOW	D_APPLI_CANL	5_43	CAN-Low3	CAN3_L	No
Lokalerr CAN-Low	D_LO_CANL	5_44	local CAN-Low	LOCAN_L	No

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Zündspule 1	A_P_ZSZ1	6_01	Ignition coil 1	IGC0	No
Zündspule 5	A_P_ZSZ2	6_02	Ignition coil 5	IGC4	No
Zündspule 3	A_P_ZSZ3	6_03	Ignition coil 3	IGC2	No
Zündspule 6	A_P_ZSZ4	6_04	Ignition coil 6	IGC5	No
Zündspule 2	A_P_ZSZ5	6_05	Ignition coil 2	IGC1	No
Zündspule 4	A_P_ZSZ6	6_06	Ignition coil 4	IGC3	No
Masse (nicht angeschlossen)	M_ZUE	6_07	GND (not connected)	IG_GND	not used
Masse (nicht angeschlossen)	M_ZUE	6_08	GND (not connected)	IG_GND	not used
Masse (nicht angeschlossen)	M_ZUE	6_09	GND (not connected)	IG_GND	not used
Masse (nicht angeschlossen)	M_ZUE	6_10	GND (not connected)	IG_GND	not used
Masse (nicht angeschlossen)	M_ZUE	6_11	GND (not connected)	IG_GND	not used
Masse (nicht angeschlossen)	M_ZUE	6_12	GND (not connected)	IG_GND	not used
Einspritzventil 1	A_P_EVZ1	7_01	Injection valve 1	IV_0	Yes
Einspritzventil 5	A_P_EVZ2	7_02	Injection valve 5	IV_4	Yes
Einspritzventil 3	A_P_EVZ3	7_03	Injection valve 3	IV_2	Yes
NTC- Wasser (Motortemperatur)	E_A_TMOT	7_04	coolant temperature	TCO	Yes
VANOS Einlass	A_T_NWE	7_05	infinitely variable valve timing inlet	IVVT_IN	Yes
Datenclock VVT Sensor	A_P_CLKS1	7_06	Data clock VVT sensor	PCLK1S1	Yes
Dateneingang Führungssensor VVT	E_T_DAT1S1	7_07	Data input main sensor VVT	TDAT1S1	Yes
Chip Select Referenzsensor VVT	A_P_CS2S1	7_08	Chip select reference sensor VVT	PCS2S1	Yes
Dateneingang Referenzsensor VVT	E_T_DAT2S1	7_09	Data input reference sensor VVT	TDAT2S1	Yes
Schirm VVT	W_VVTS1	7_10	Shield VVT	VVT_SHIELD	Yes
Nockenwellengeber Einlaß	E_P_NWGE	7_11	camshaft position sensor inlet	CAM_IN	Yes
Nockenwellengeber Auslaß	E_P_NWGA	7_12	camshaft position sensor exhaust	CAM_EX	Yes
Öldruck	E_S_OLD	7_13	oil pressure	POIL	No
Einspritzventil 6	A_P_EVZ4	7_14	Injection valve 6	IV_5	Yes
Einspritzventil 2	A_P_EVZ5	7_15	Injection valve 2	IV_1	Yes
Einspritzventil 4	A_P_EVZ6	7_16	Injection valve 4	IV_3	Yes
Masse Motortemperaturfühler	M_TMOT	7_17	ground coolant temperature sensor	TCO_GND	Yes
Vanos Auslass	A_T_NWA	7_18	infinitely variable valve timing exhaust	IVVT_EX	Yes
Elektr. Geregelter Thermostat	A_S_KFK	7_19	el. controlled thermostat	ECT	Yes
Masse VVT-Sensor	M_VVTS1	7_20	Ground variable valve timing	VVTS1_GND	Yes
Spannungsversorgung 5V (VVT-Sensor)	A_U_VVTS1	7_21	supply voltage to VVT sensor	VVTS1_VCC	Yes
Chip Select Führungssensor VVT	A_P_CS1S1	7_22	Chip select main sensor VVT	PCS1S1	Yes
Schalt signal VVT Relais	A_S_VVTR1	7_23	VVT relay	RLY_VVT	Yes
Masse Nockenwellengeber 1 Einlaß	M_NWGE	7_24	ground camshaft position sensor inlet 1	CAM_IN_GND	Yes
Masse Nockenwellengeber 1 Auslaß	M_NWGA	7_25	ground camshaft position sensor exhaust 1	CAM_EX_GND	Yes
Generatorschnittstelle	D_BSD	7_26	generator interface	BSD	No



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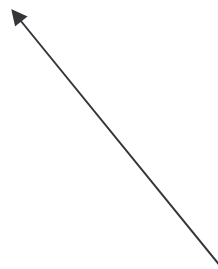
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9.21 Location of the Data Link Connector

9.21.1 Location of the Data Link Connector for models X5 3.0i



Position OBD II Connector

9.22 Drawing and Location of the Malfunction Indicator Light

9.22.1 Drawing and Location of the Malfunction Indicator Light for model X5 3.0i



**Malfunction Indicator
Light**

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9.23 Calculated load and fuel trim determination

The calculated engine load "LOAD_CLC [%]" is based on the measured mass air flow (metered by the hot-film air-mass sensor (HFM)).

Strategy:

A 2-dimensional map is used to interpolate the calculated engine load "LOAD_CLC [%]" depending on metered mass air flow and engine speed. A weighting factor is applied to compensate the altitude influence.

The calculation is performed as follows:

$$\text{LOAD_CLC [\%]} = \text{LOAD_CLC_RAW} \cdot f(\text{metered mass air flow, engine speed}) \cdot (1013\text{hPa} / \text{ambient pressure}) \cdot 100\%$$

with:

LOAD_CLC	calculated engine load in % with altitude correction
LOAD_CLC_RAW	calculated engine load in % without altitude correction

In case of a malfunction of the HFM, the metered mass air flow is substituted by a modeled mass air flow value.



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9.24 Start Emission Reduction Strategy Monitoring

P1561,1562

All parameters, that are relevant during the cat heating phase, are monitored by standard monitoring functions:

Relevant Components during Cat Heating Phase						Impact of faulty Component on Cat Heating Parameter				
Comp./ System	Parameter	needed for	Component Diagnosis	Diagnosis during Cat Heating	Emission Impact > 1,5xGW	Idle Speed	Ignition Angle	Engine Lambda	Transmission Shifting Point	Camshaft Position
Secondary Air	Secondary Air Mass	Enleanment exhaust gas	Secondary Air diagnosis	Yes	Yes, dep. to variant and emission class			X		
Injection Valve	Injection time	Enrichment lambda_engine < 1 Enleanment lambda_engine > 1	Output stage diagnosis	Yes	none	X	X	X		
			Misfire detection	Yes						
			Fuel supply diagnosis	no						
Mass Air Flow Sensor	Air Mass - Input for maps	Larger overlap, VANOS End position	Air mass flow sensor diagnosis	Yes	none	X	X	X		
			Air mass model diagnosis	Yes						
			Fuel supply diagnosis	no						
Throttle Position	Angle	Mass Air Flow	Power stage, accelerator pedal diagnosis	Yes	none	X	X	X		
Valvetronic	Valve Lift	Load-control	Valvetronic electrical / mechanical diagnosis	Yes	none	X	X	X		
			Air mass model diagnosis	Yes						
Phase Sensor (Camshaft)	Valve overlap	Larger overlap	RPM sensor diagnosis	Yes	none					X
Camshaft Position Actuator	Valve overlap	Larger overlap	Output Stage diagnosis	Yes	none					X
			Camshaft position actuator	Yes						
RPM Sensor	Engine Speed	Idle speed increase	RPM sensor diagnosis	Yes	none	X				
Idle Speed	Engine Speed	EVAP, Camshaft Position Shifting point	Idle speed diagnosis	Yes	none	X				X
CAN-Communication with Transmission	CAN-bus	Shifting point	Timeout CAN-message	Yes	none	X			X	
Coolant Temperature Sensor	Temperature	Input for maps	Electrical plausibility	Yes	Yes	X	X	X	X	X
			Stuck signal	no						
ECM	Signals	Calculation	Self Check RAM,ROM,VW-dog	Yes	- - -					
Ignition	Ignition angle	Optimum: retarded ignition	Misfire detection	Yes	Yes, dep. to emission class		X			

Illustration 1 (Standard monitoring functions during Cat Heating - Overview)



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To fulfill the legal requirements, the monitoring of the idle speed is now extended to the cold start phase. In case of an error, the specific P-Codes:

- P1561 Cold Start Idle Air Control System RPM lower than expected
- P1562 Cold Start Idle Air Control System RPM higher than expected

are set.

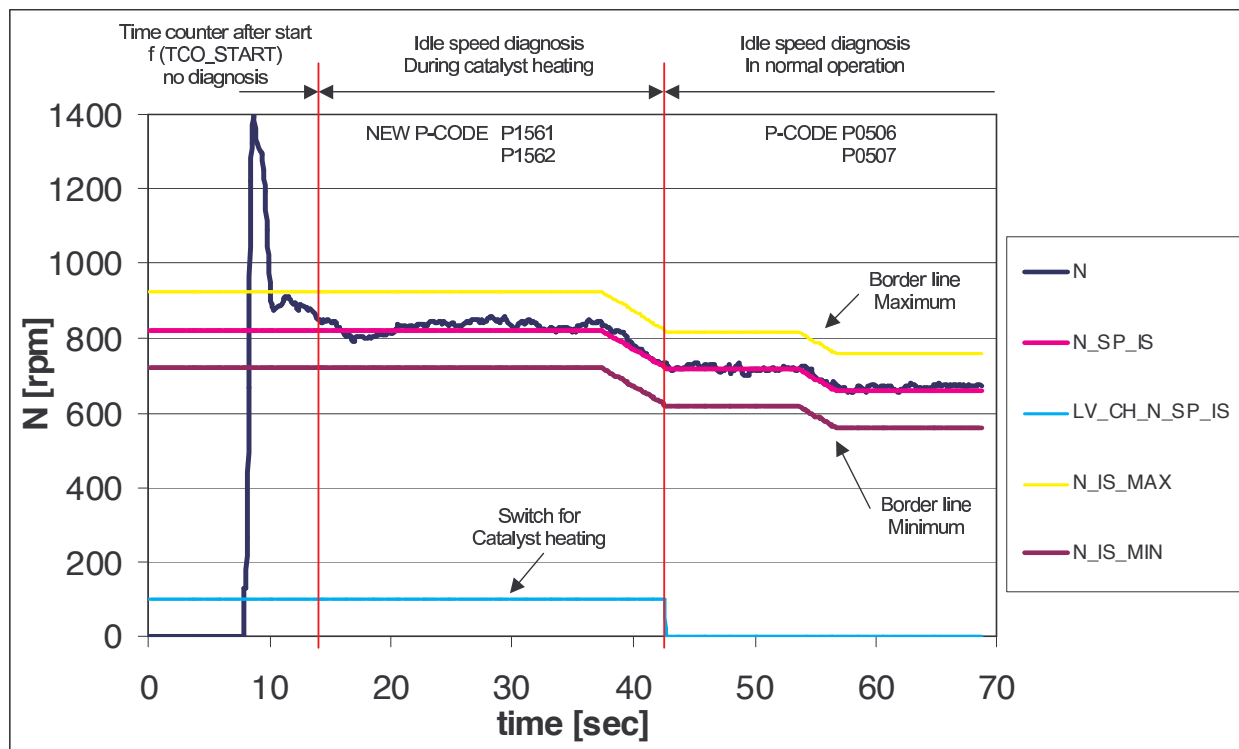


Illustration 2 (Idle speed control)



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During cat heating, it is essential to make sure, that enough thermal energy is applied to the catalyst to heat it up as quick as possible.

Therefore it is target to limit the ignition timing to the earliest possible value during the cat heating phase.

If there would be a demand for more torque and therefore for an advanced ignition timing beyond the limits, the engine would be allowed to stall instead of fulfilling the demand.

The torque limits are calibrated the way, that the emissions stay below 1.5 times of the limits.

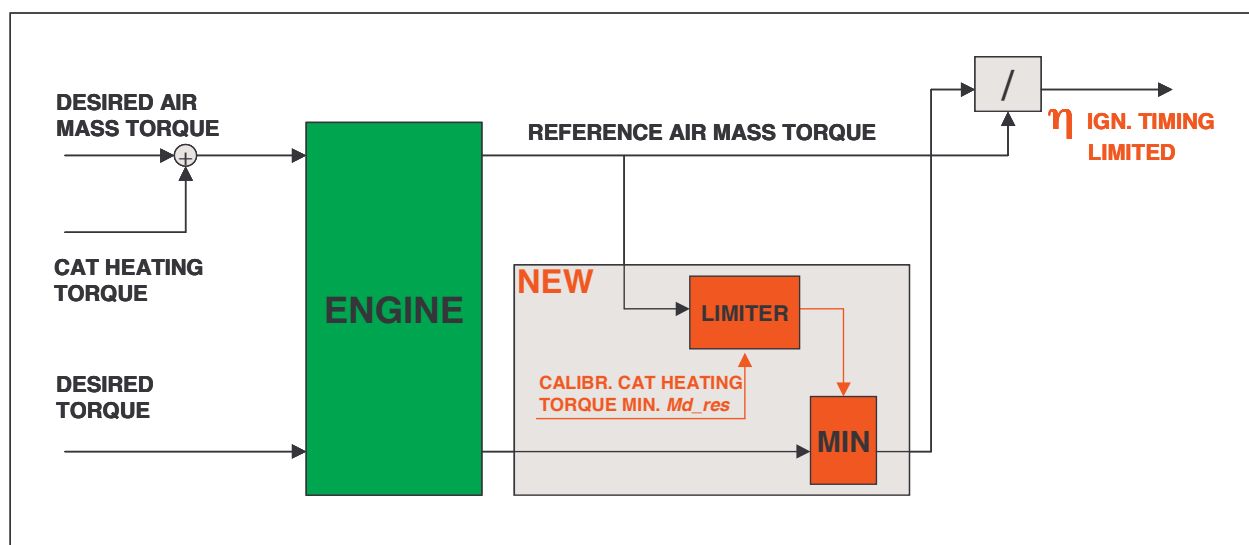


Illustration 3 (Flowchart)



Known System:

During normal driving, the ignition timing desired torque corresponds to the air mass desired torque, which determines the ignition timing. During the cat heating phase, the cat heating torque is added to the air mass desired torque, resulting in a higher reference air mass torque. The efficiency, desired torque divided by the reference torque, determines the ignition timing.

New System (BMW-development):

The earliest possible ignition timing is determined by the limitation of the torque reserve to a minimum value during the cat heating phase. For this, the required minimum cat heating torque is subtracted from the reference air mass torque. The thus reduced efficiency leads to a safe ignition retard and limits the ignition timing during the cat heating measures.

Limitation of ignition timing to the earliest possible ignition timing during the cat heating phase by limitation of torque reserve to the minimum required torque reserve.

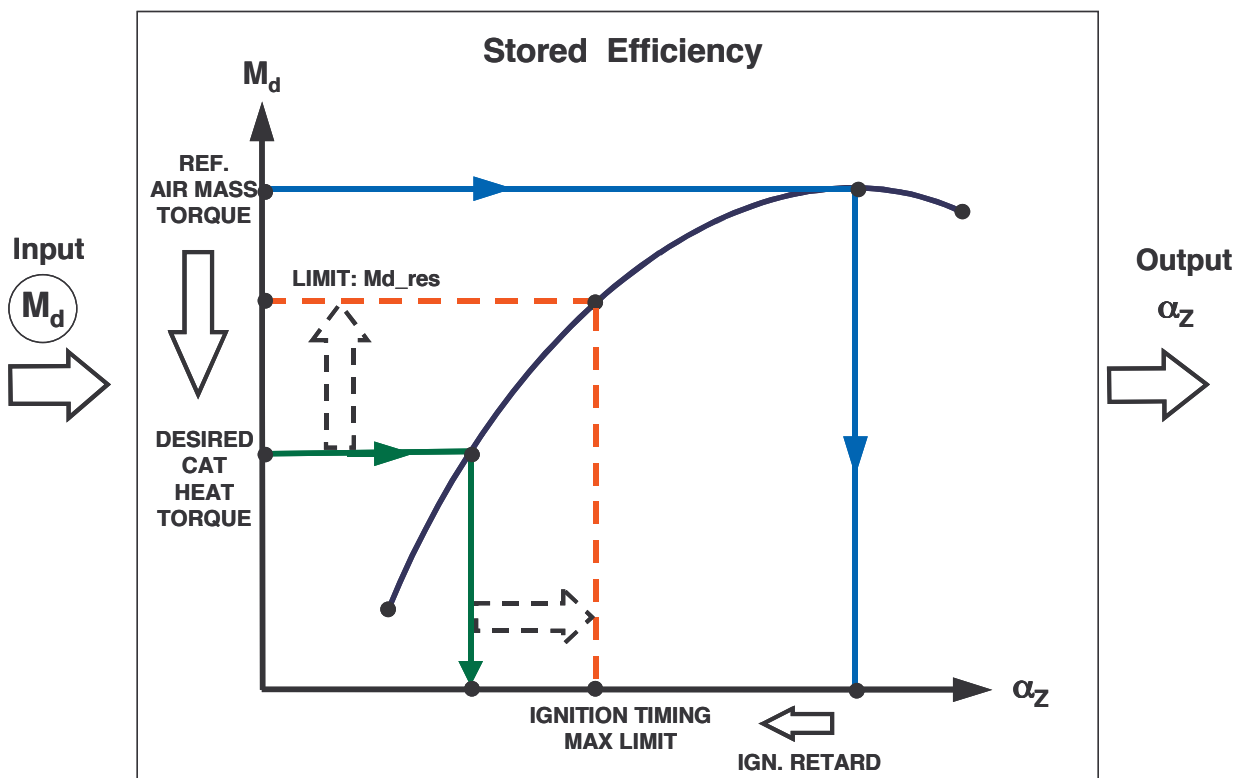


Illustration 4 (Diagram torque characteristic line and ignition timing)



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The maximum ignition timing after cold start with new BMW method:

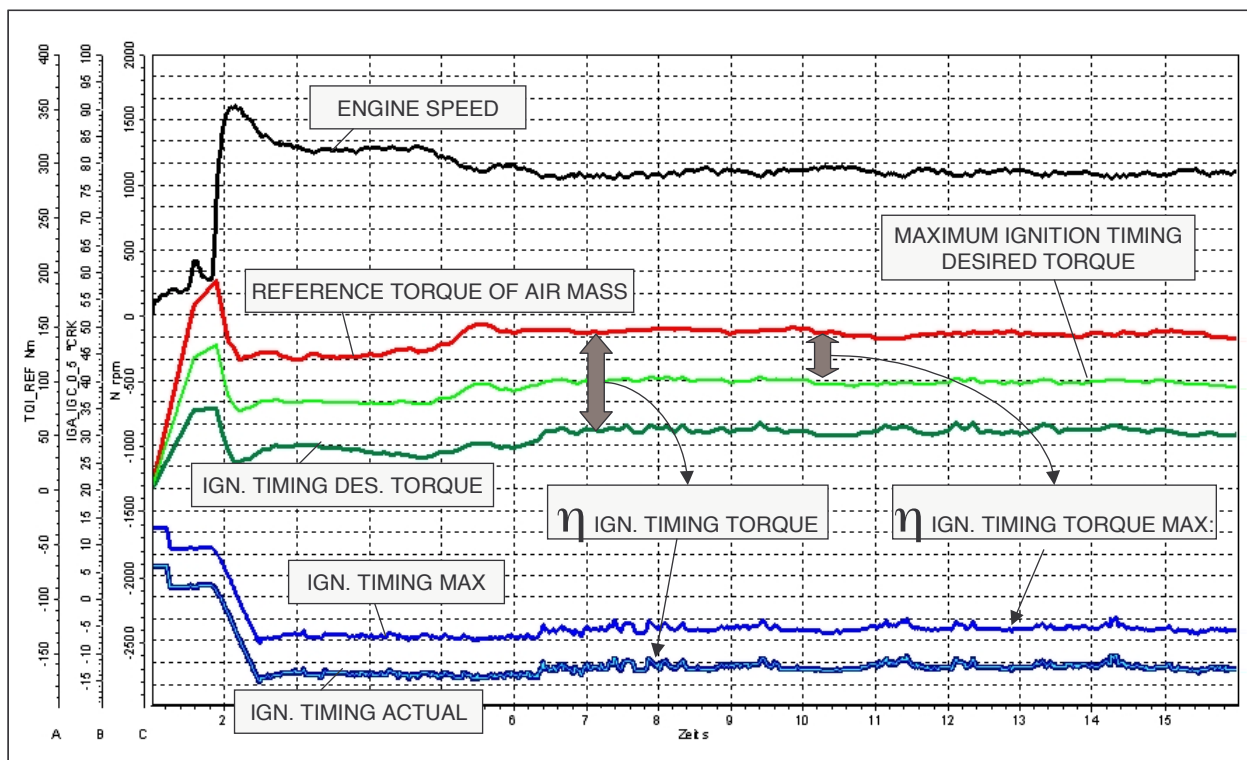


Illustration 5 (Measuring Data)